

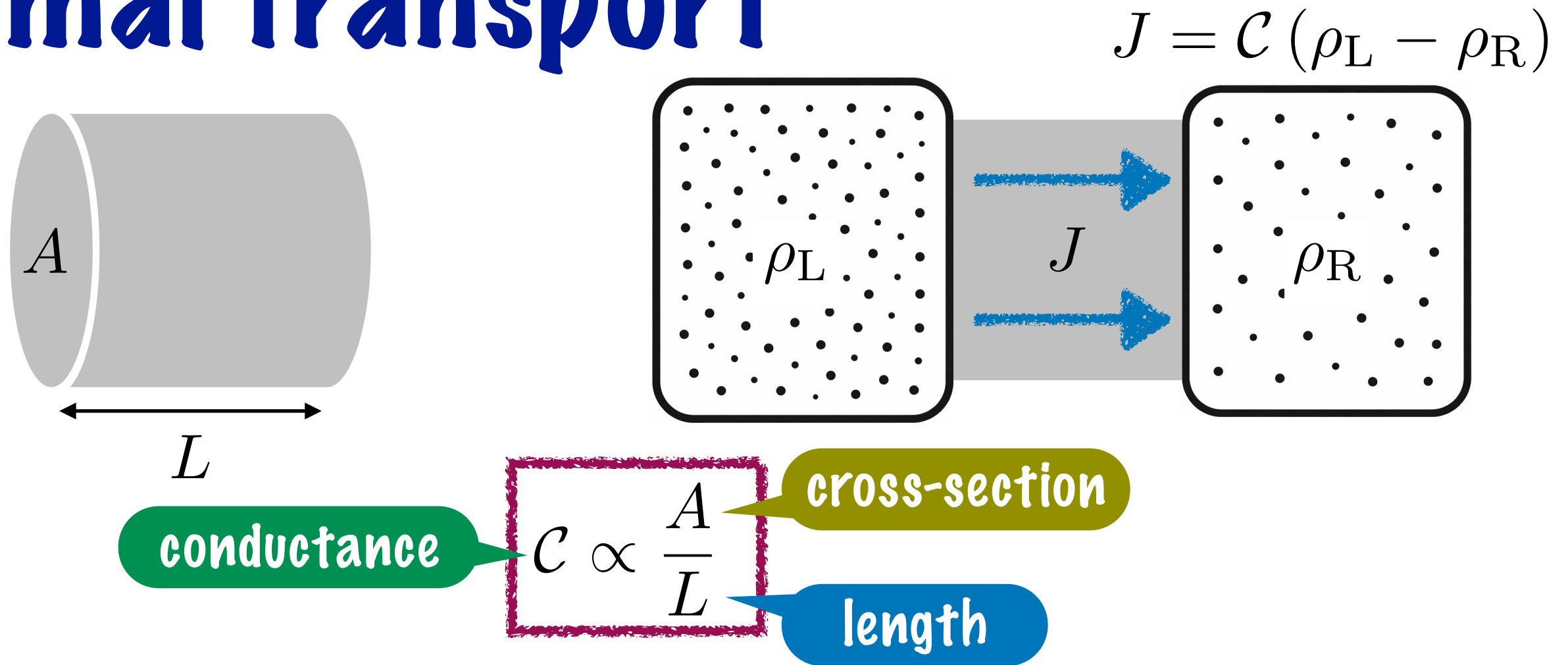


Hierarchical Lorentz Mirror Model Normal Transport and a Universal $2/3$ Mean-Variance Law

Raphael Lefevere and Hal Tasaki

webinar @ YouTube: 2026/2/3

normal transport



follows from diffusive (Ohm's, Fick's, or Fourier's) laws

do we get normal transport from fully deterministic (and non-chaotic) dynamics in a quenched random environment?

Lorentz gas, especially its discrete version,
the Lorentz mirror model

Lorentz 1905, Ruijgrok, Cohen 1988

Lorentz mirror model

Hierarchical model

definition and recursion relation

main theorem on normal transport

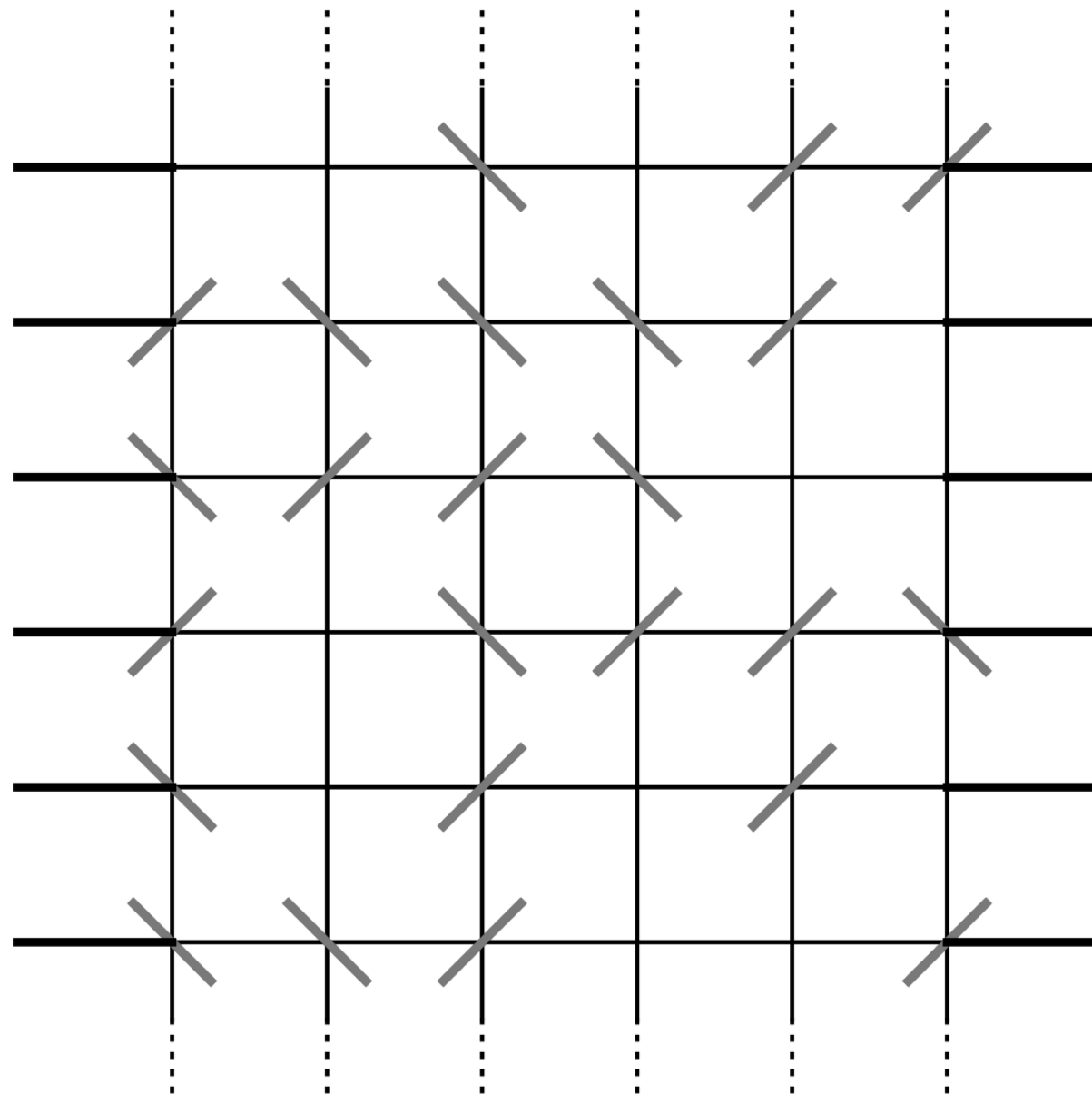
Gaussian closure and the “2/3 law”

Lorentz mirror model in $d = 3$: numerics

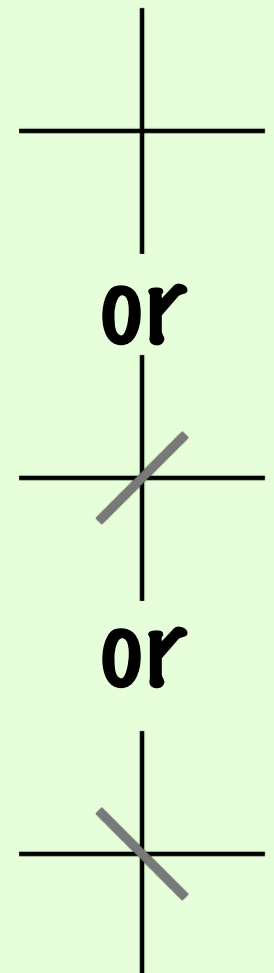
summary

appendix: brief ideas of the proof

Lorentz mirror model ($d=2$)

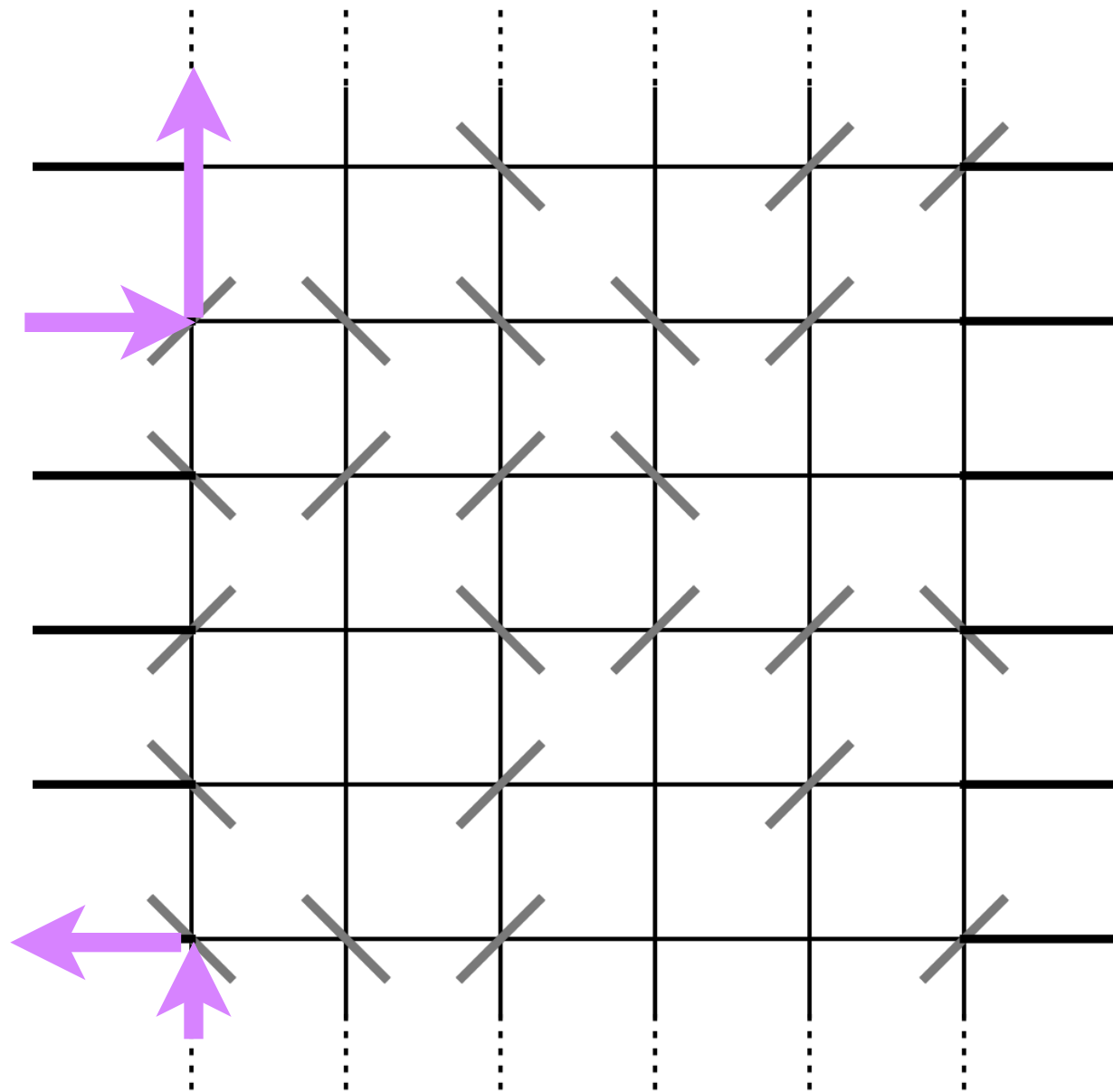


with
probability $1/3$



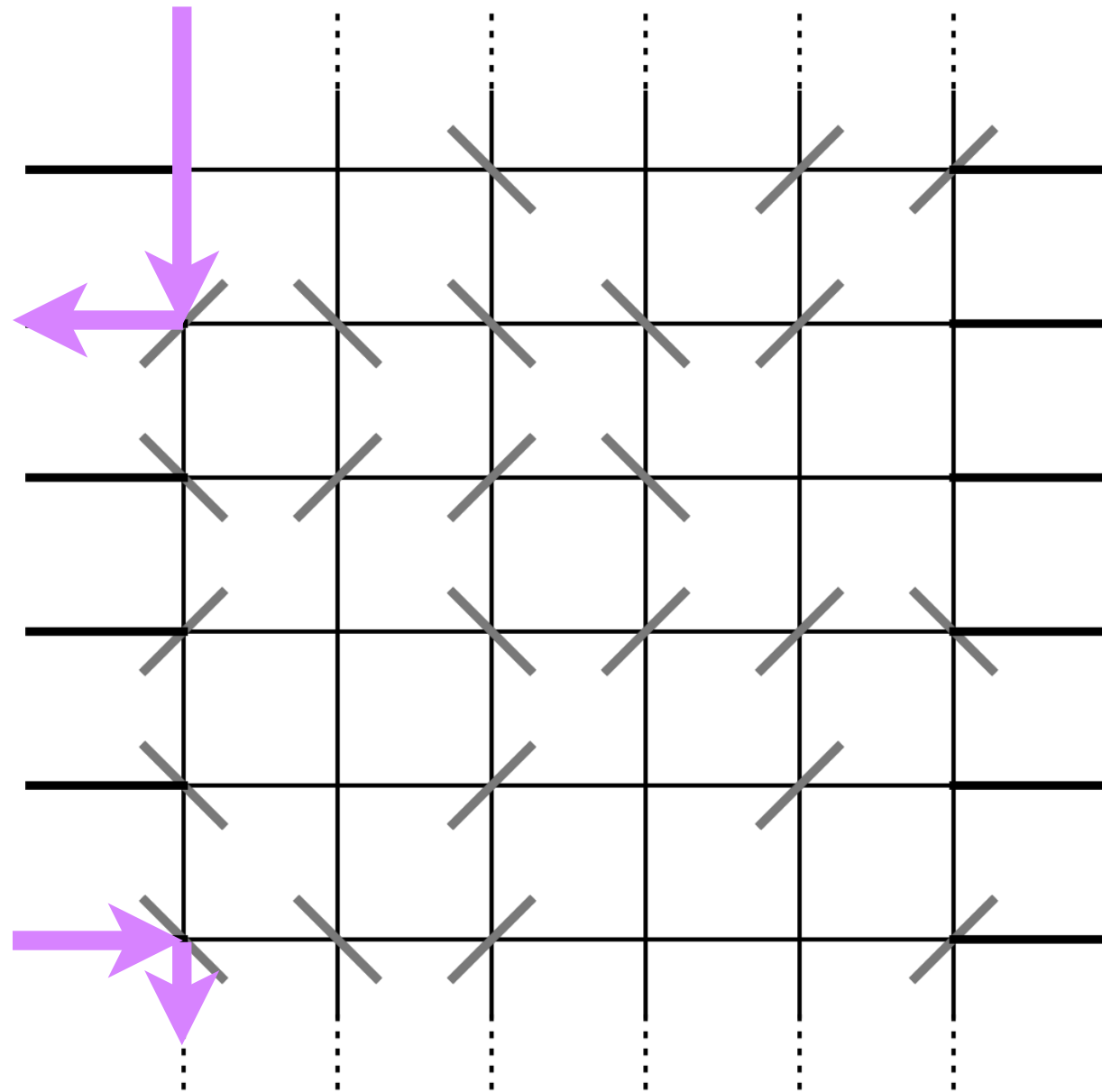
$L \times L$ square lattice
horizontal: open b.c., vertical: periodic b.c
add L external edges on the left and the right
place mirrors randomly

Lorentz mirror model ($d=2$)



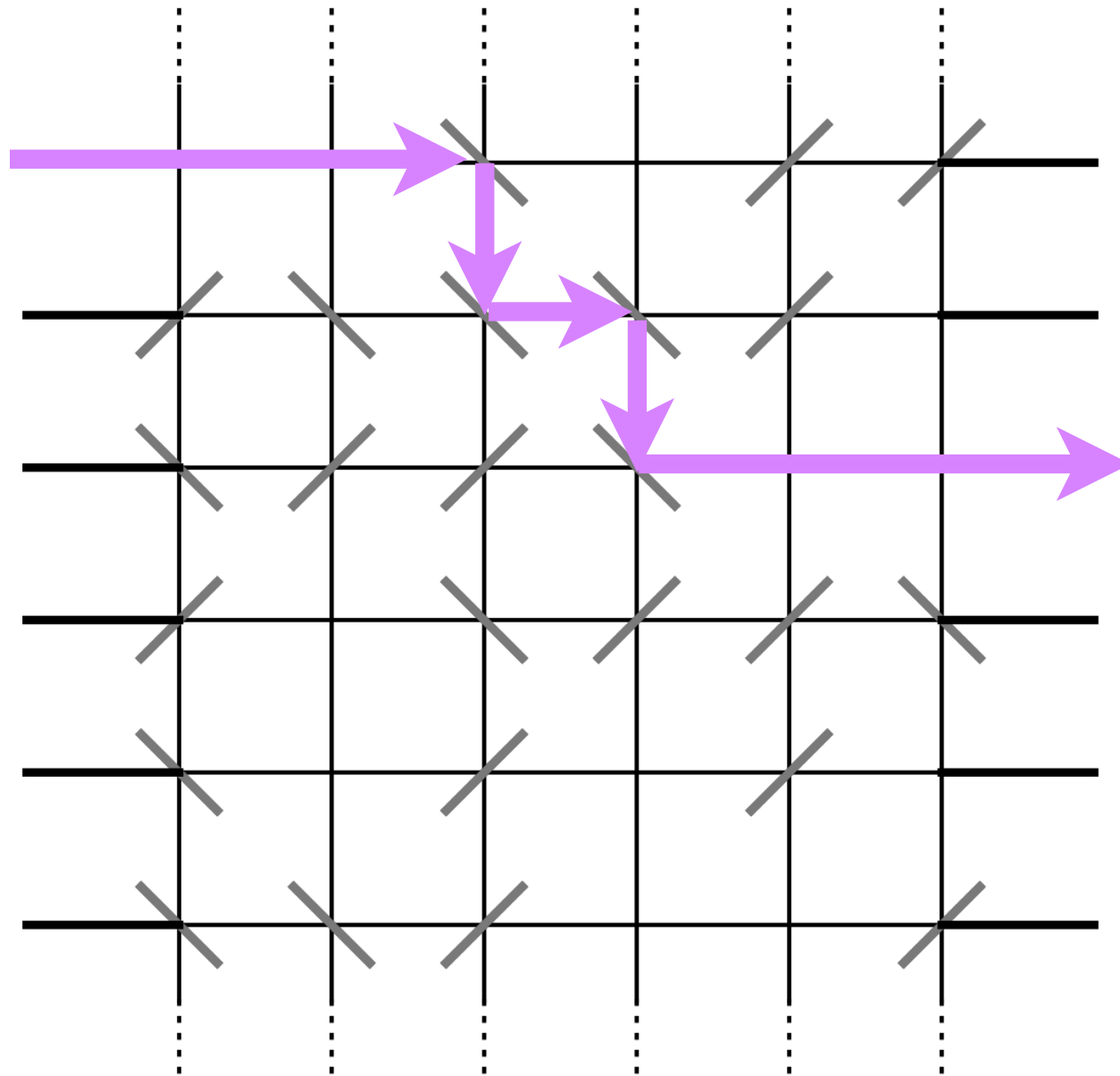
a particle (or light ray) is reflected by mirrors

Lorentz mirror model ($d=2$)



a particle (or light ray) is reflected by mirrors
trajectory is reversible

Lorentz mirror model ($d=2$)



a particle (or light ray) is reflected by mirrors
trajectory is reversible
looks like a random walk

non-backtracking random walk

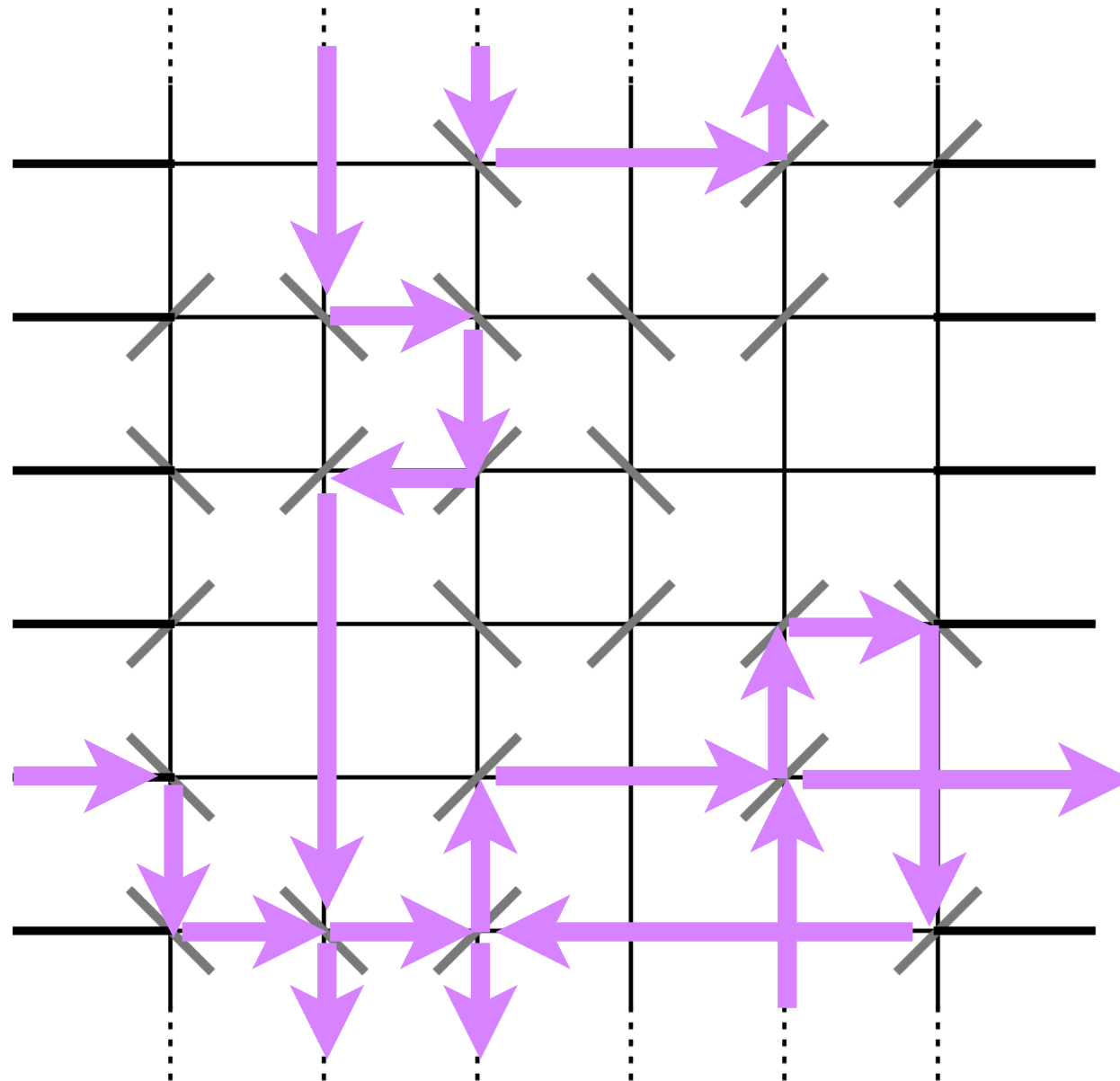
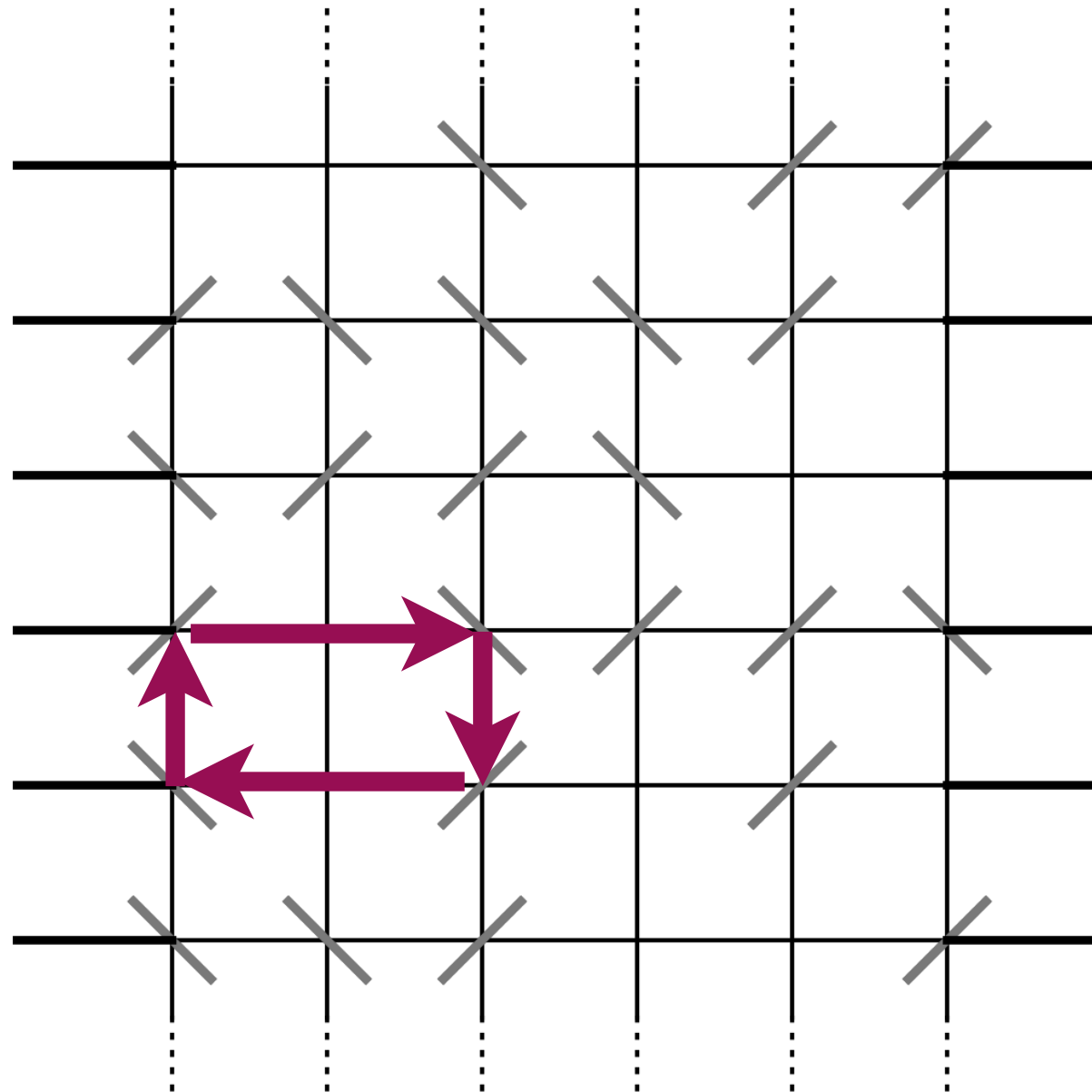


illustration by John Tenniel

Lorentz mirror model ($d=2$)

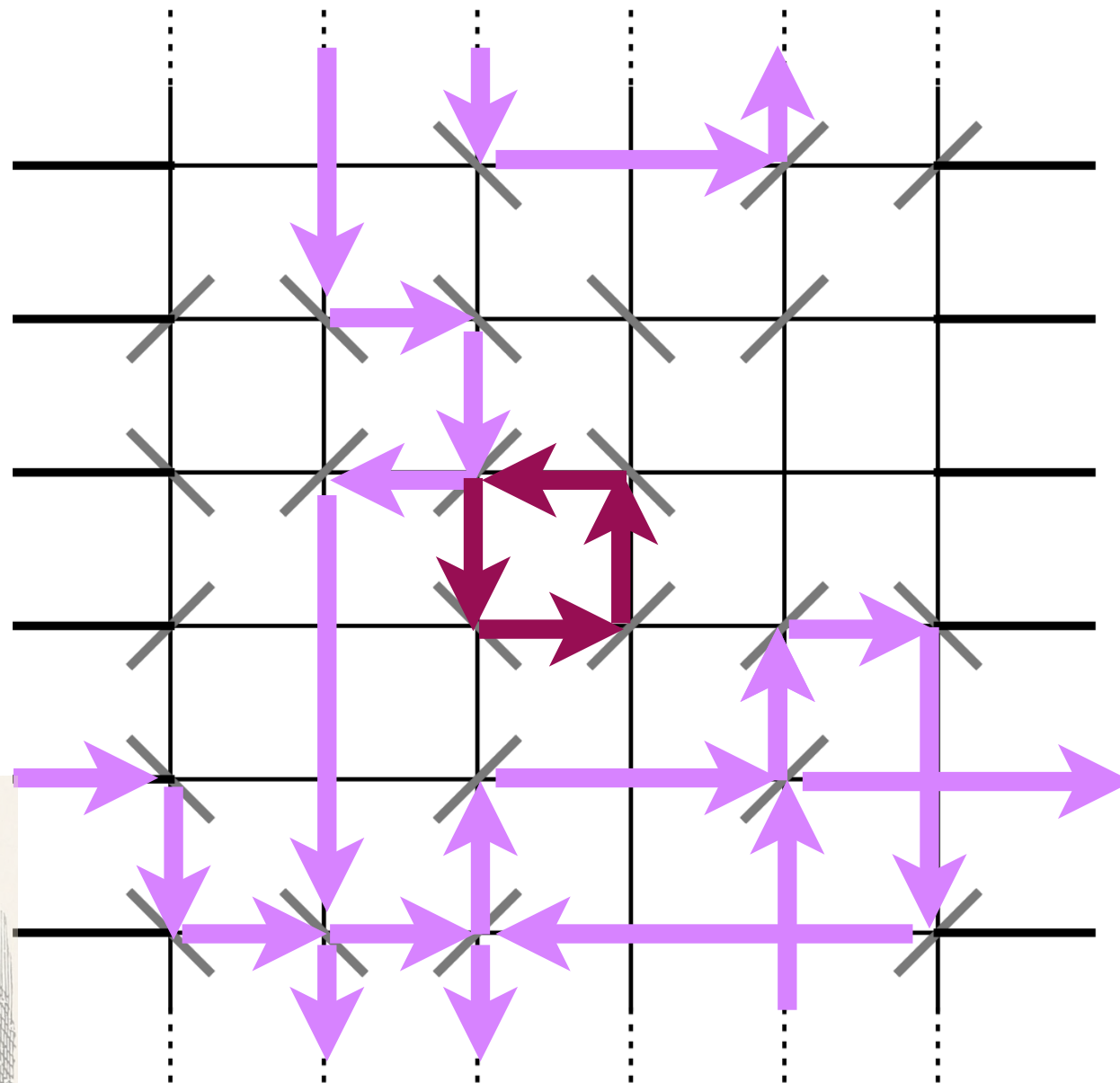


a particle (or light ray) is reflected by mirrors
trajectory is reversible

looks like a random walk
but has strong memory!

many loops!

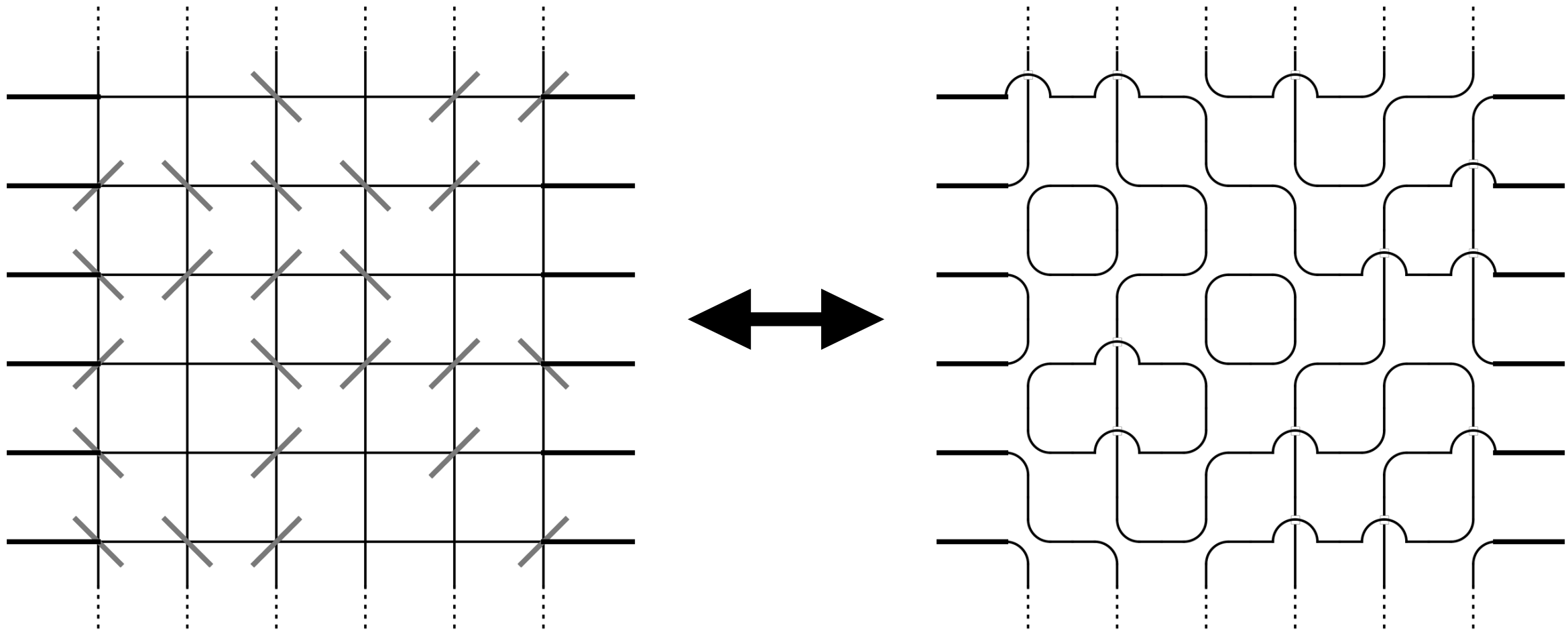
Lorentz mirror model ($d=2$)



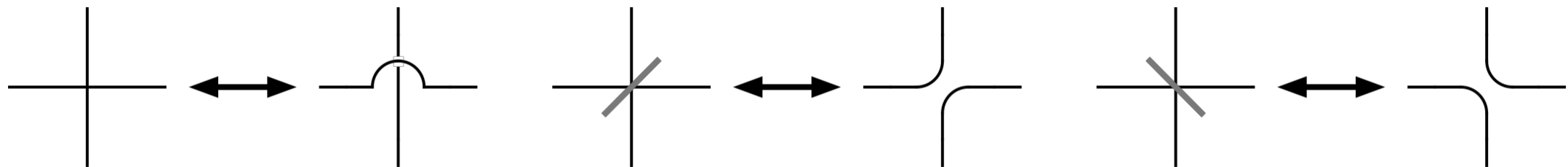
although there are many loops inside, a particle injected through an external edge must exit through another external edge (either on left or right)

no traps!!

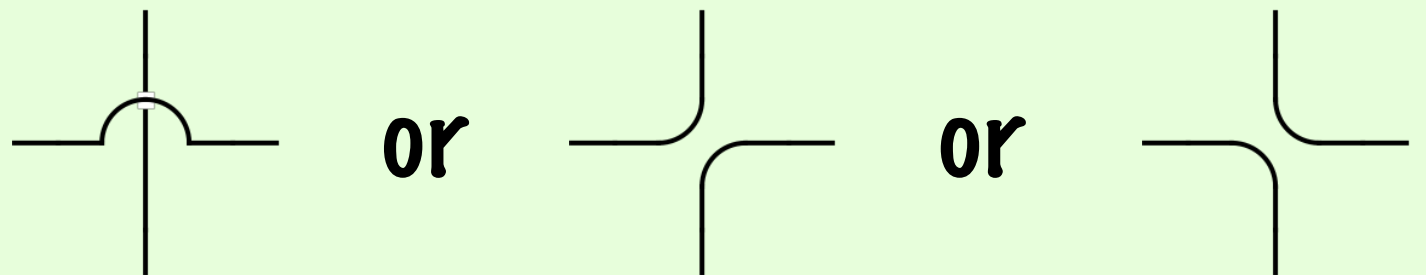
from mirrors to local pairings



mirrors (and their absence) are equivalent to local pairings

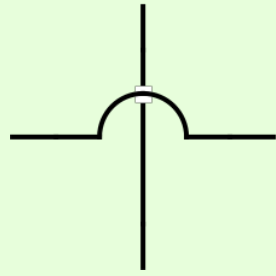


with probability $1/3$

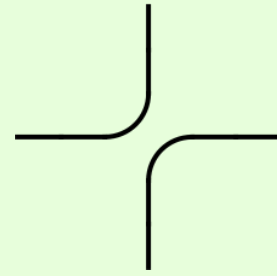


matching of external edges, and crossings

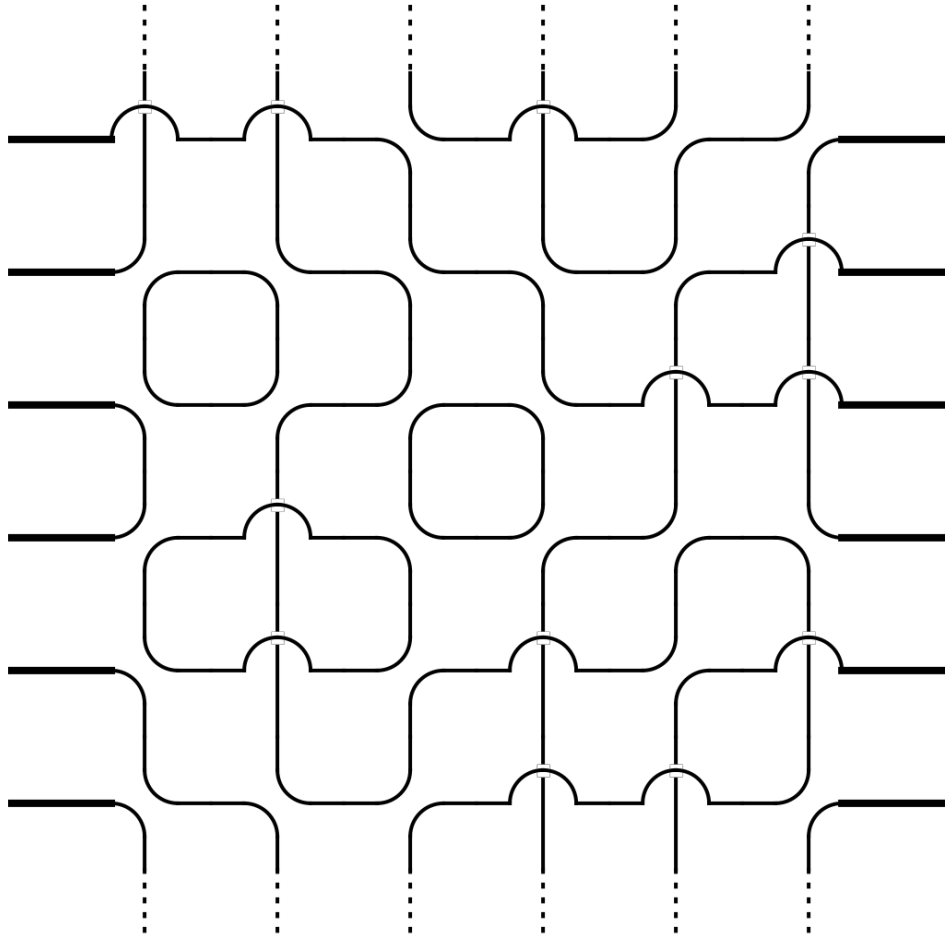
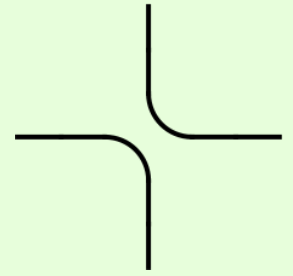
with probability $1/3$



or

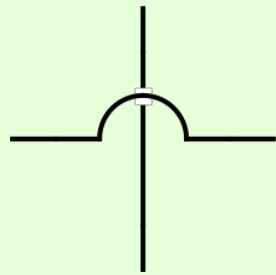


or

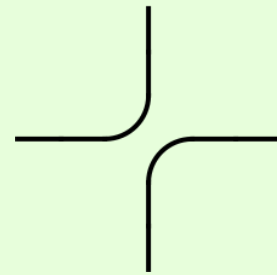


matching of external edges, and crossings

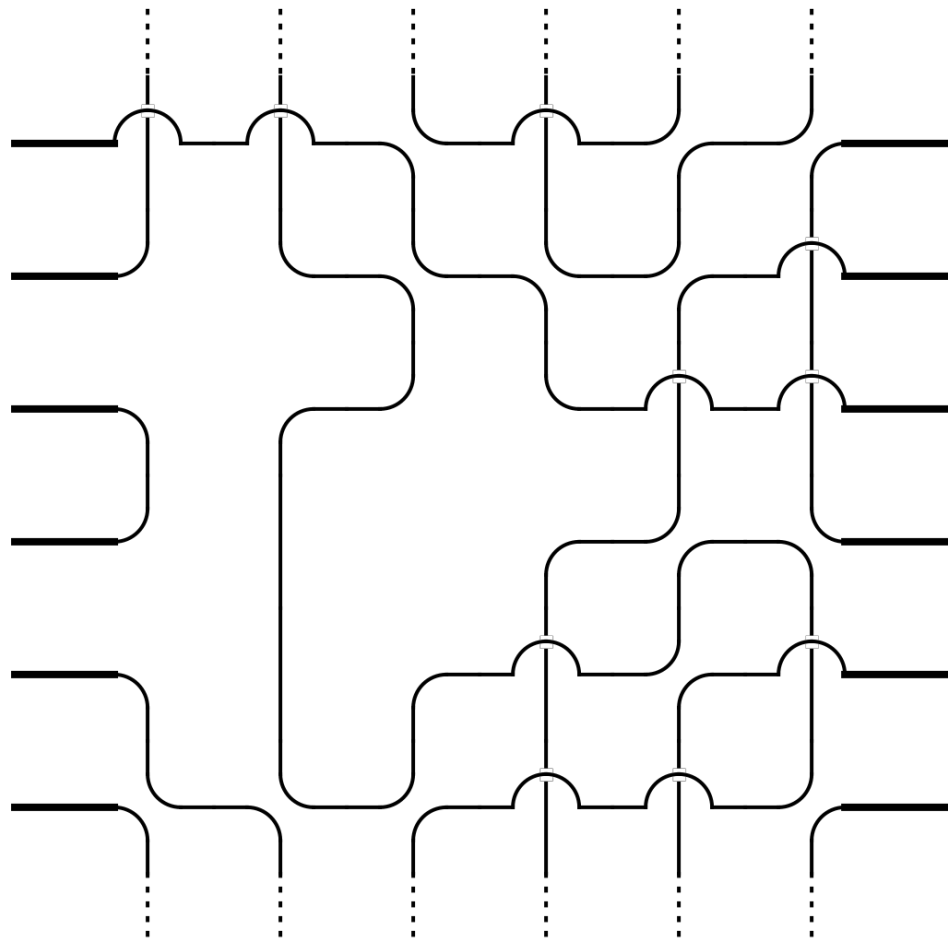
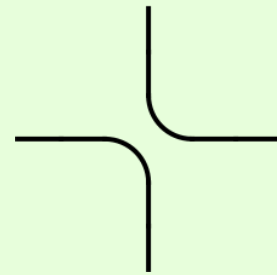
with probability $1/3$



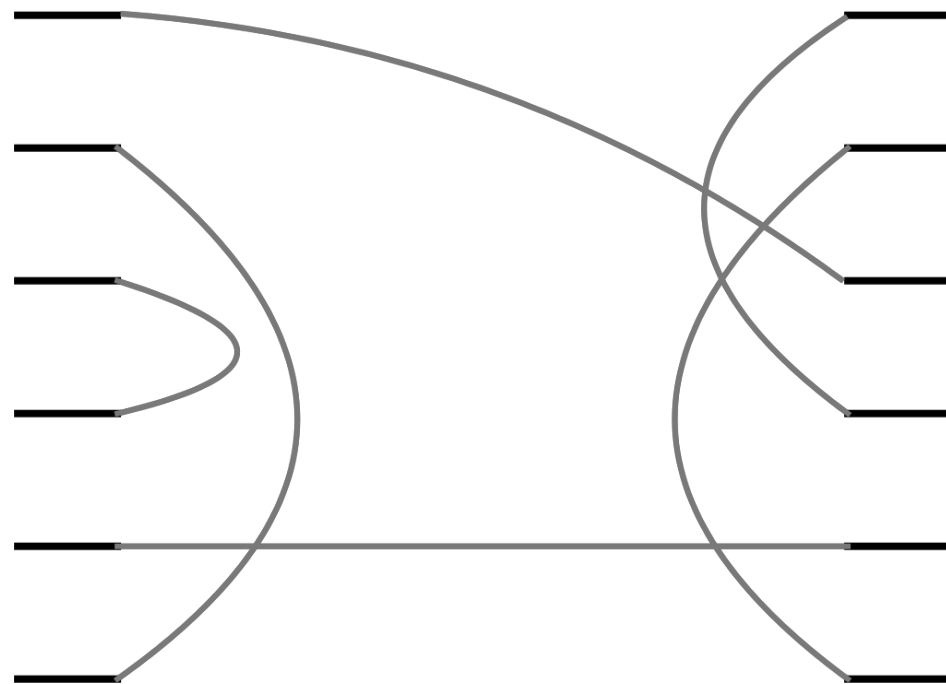
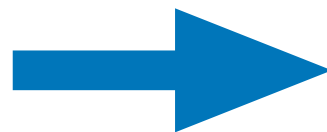
or



or



remove loops



a random perfect matching
(pairing) of external edges

left-right pair = crossing

& the number of crossings

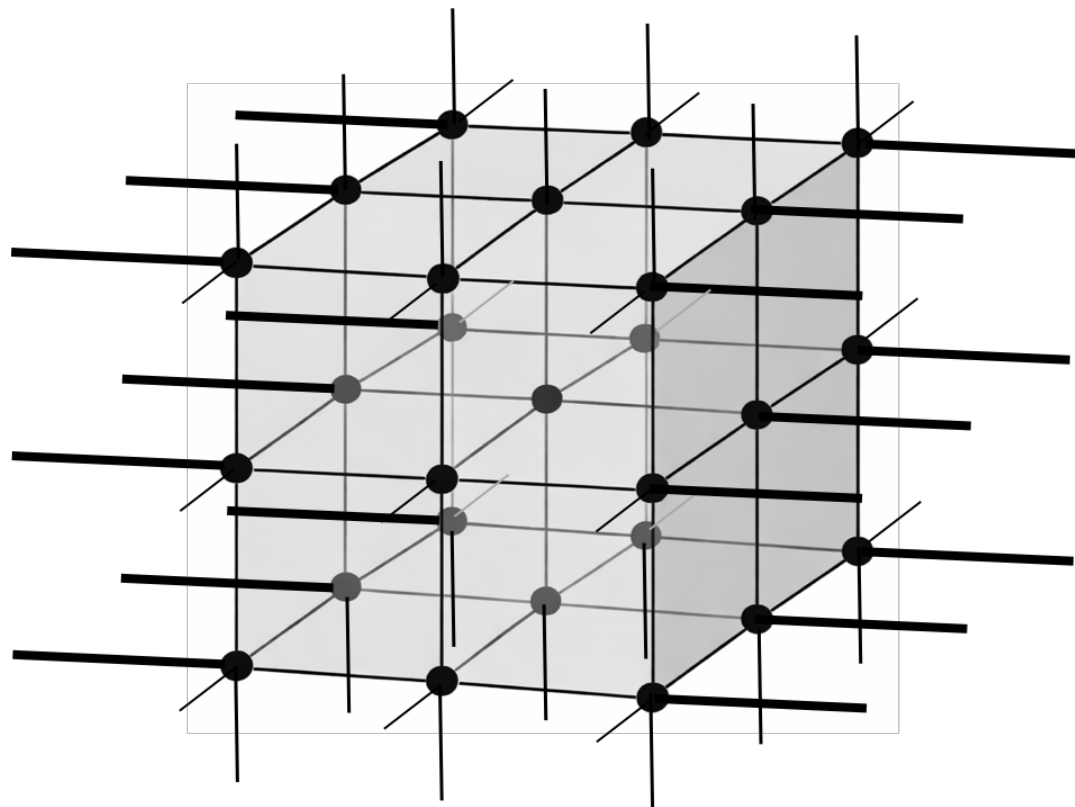
conductance

Lorentz mirror model in $d \geq 2$

d dimensional $L \times \dots \times L$ hyper cubic lattice

horizontal: open b.c., other directions: periodic b.c

add L^{d-1} external edges on the left and the right



Lorentz mirror model in $d \geq 2$

d dimensional $L \times \dots \times L$ hyper cubic lattice

horizontal: open b.c., other directions: periodic b.c

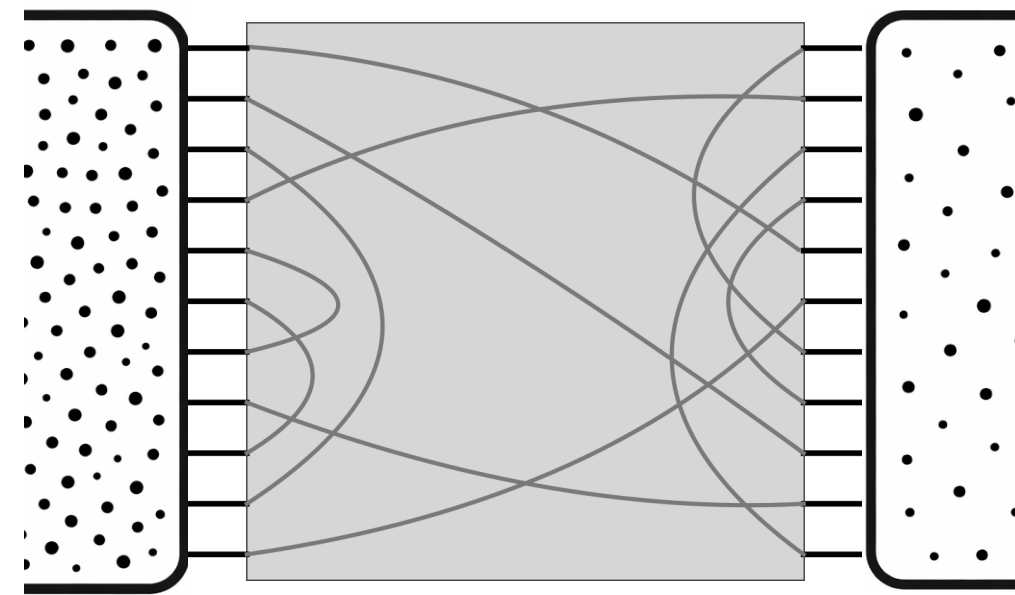
add L^{d-1} external edges on the left and the right

at each vertex, independently choose one of the $(2d - 1)!!$ pairings of the $2d$ incident edges with equal probability

local pairings induce random matching of the $2L^{d-1}$ external edges

conductance $\mathcal{C}$: the number of crossings (matched pairs of left and right external edges)

when left and right boundaries are exposed to particle baths with different densities, the stationary current is proportional to $\mathcal{C}$



normal transport in the Lorentz mirror model

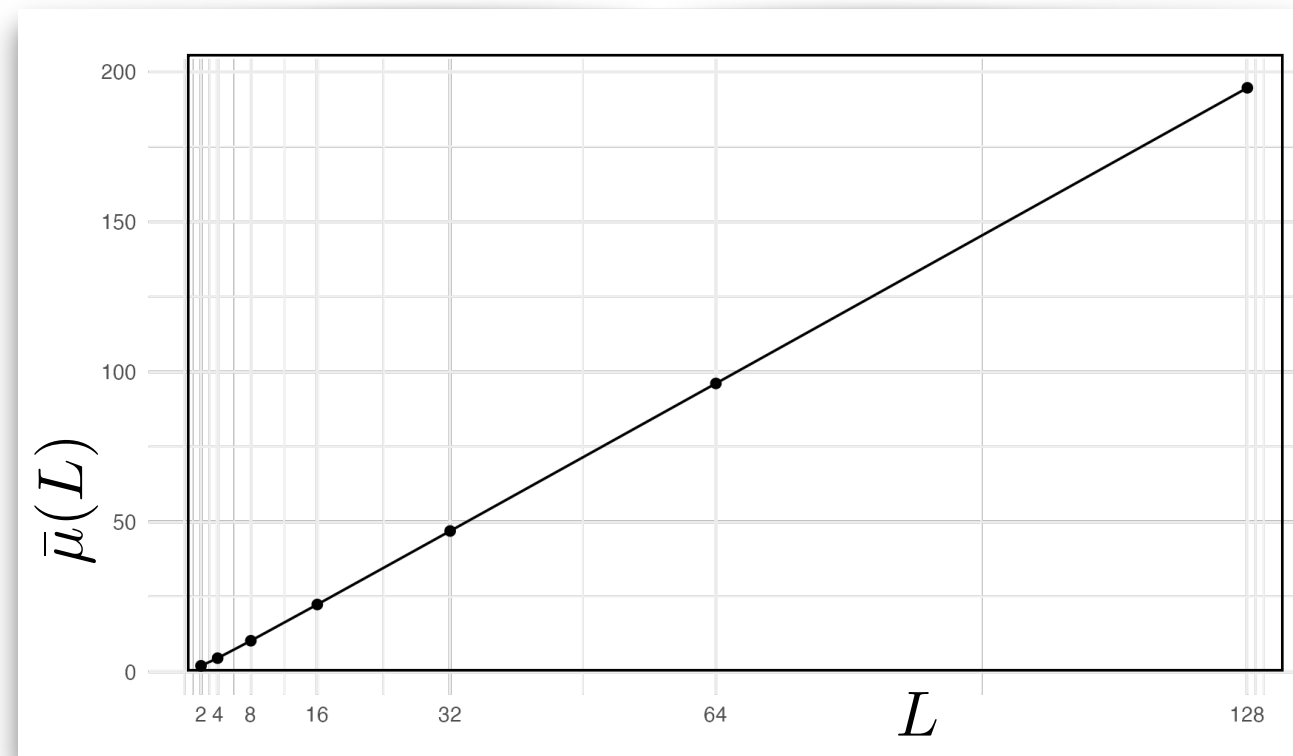
mean conductance $\bar{\mu}(L) = \langle \mathcal{C} \rangle$

normal transport

$$\bar{\mu}(L) \propto \frac{A}{L} = \frac{L^{d-1}}{L} = L^{d-2}$$

numerical and theoretical
support for normal
transport in $d = 3$

Chiffaudel, Lefevere 2016, Lefevere 2025



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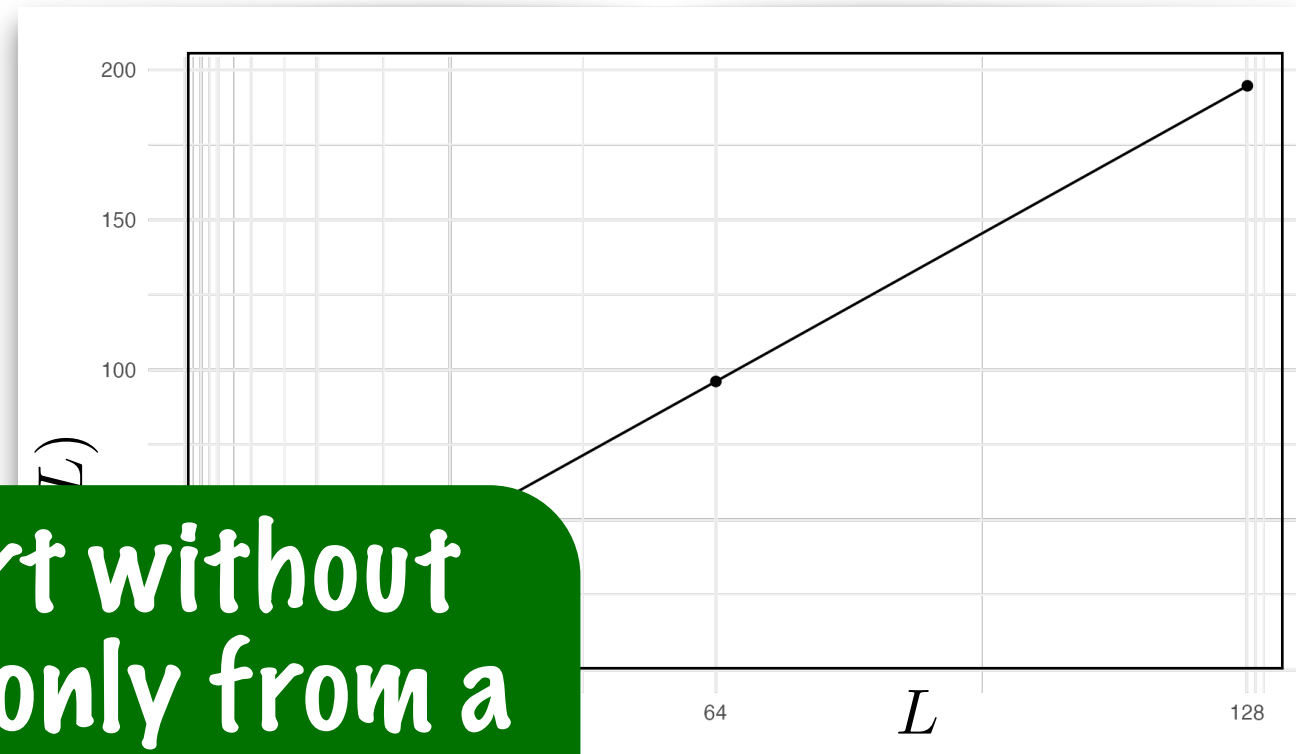
numerical and theoretical
support for normal
transport in $d = 3$

Chiffaudel, Lefevere 2016, Lefevere 2025

normal (diffusive) transport without
stochastic forcing or chaos, only from a
quenched random environment!!

the structure of the trajectories is far from diffusive!
rigorous proof seems formidable...

tractable, yet nontrivial, hierarchical version of the model



Lorentz mirror model

Hierarchical model

definition and recursion relation

main theorem on normal transport

Gaussian closure and the “2/3 law”

Lorentz mirror model in $d = 3$: numerics

summary

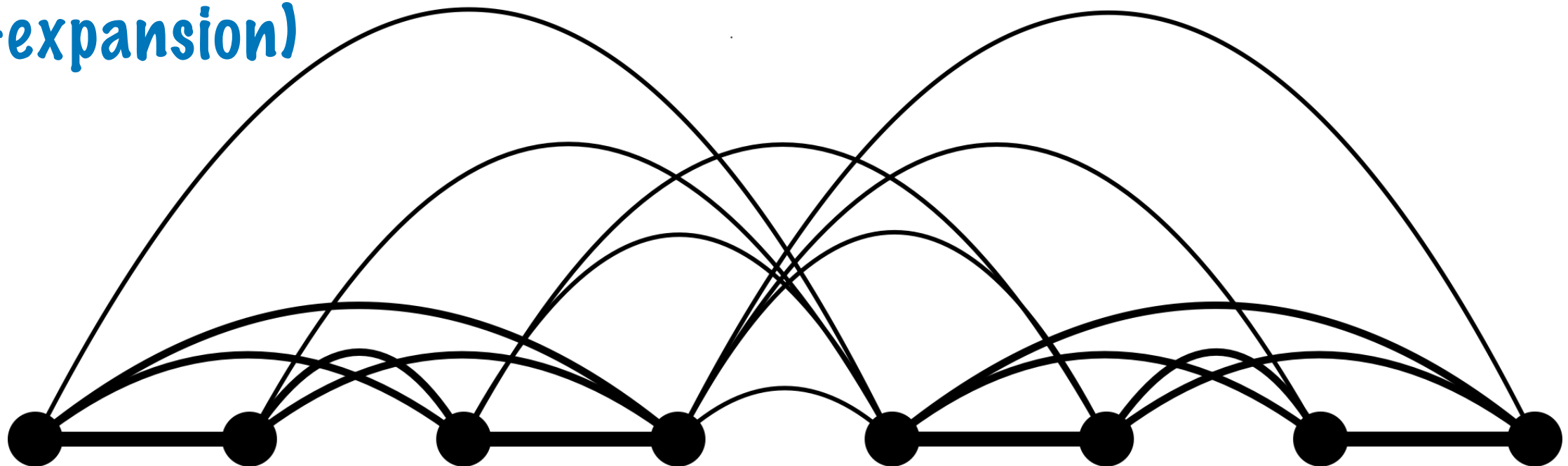
appendix: brief ideas of the proof

hierarchical models in statistical mechanics

Dyson 1969, ...

Recently Dyson introduced so-called hierarchical models imitating in many respects the lattice systems with pairwise long-range power interaction. (Bleher and Sinai, 1973)

- 📌 **mean-field-like; interactions are averaged over blocks**
 - ★ still preserves the correct dimensional scaling through an explicit hierarchy of length scales
 - ★ predicts the correct upper critical dimension, and also nontrivial critical exponents (to the leading order in the ε -expansion)



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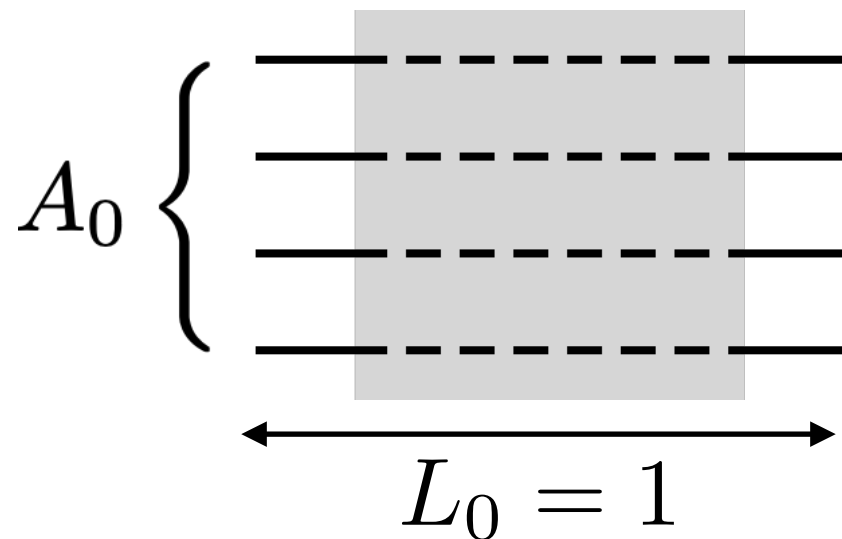
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basic setting

dimension $d = 1, 2, 3, \dots$ A_0 a positive even integer

generation $n = 0, 1, 2, \dots$

generation-0 block



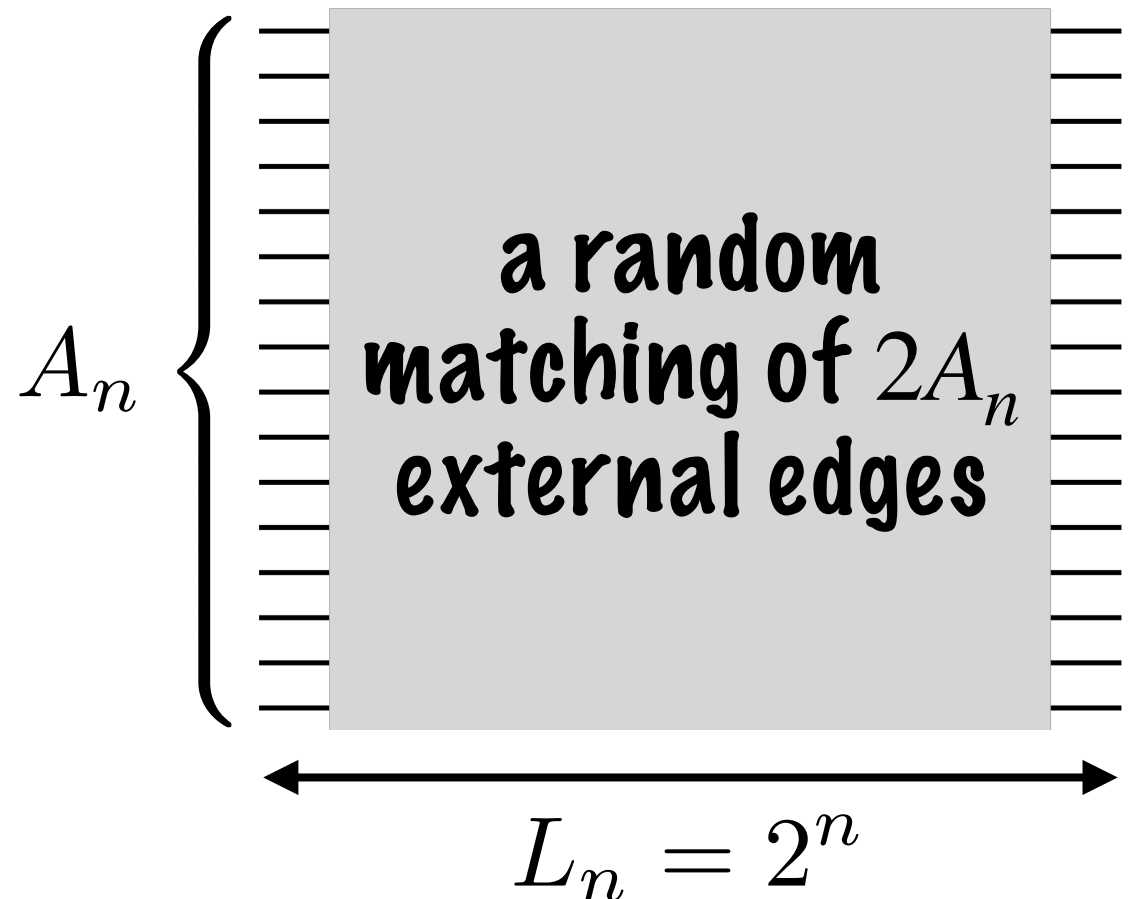
A_0 external edges on the left
and right

exactly A_0 crossings

generation- n block

A_n external edges on
the left and right

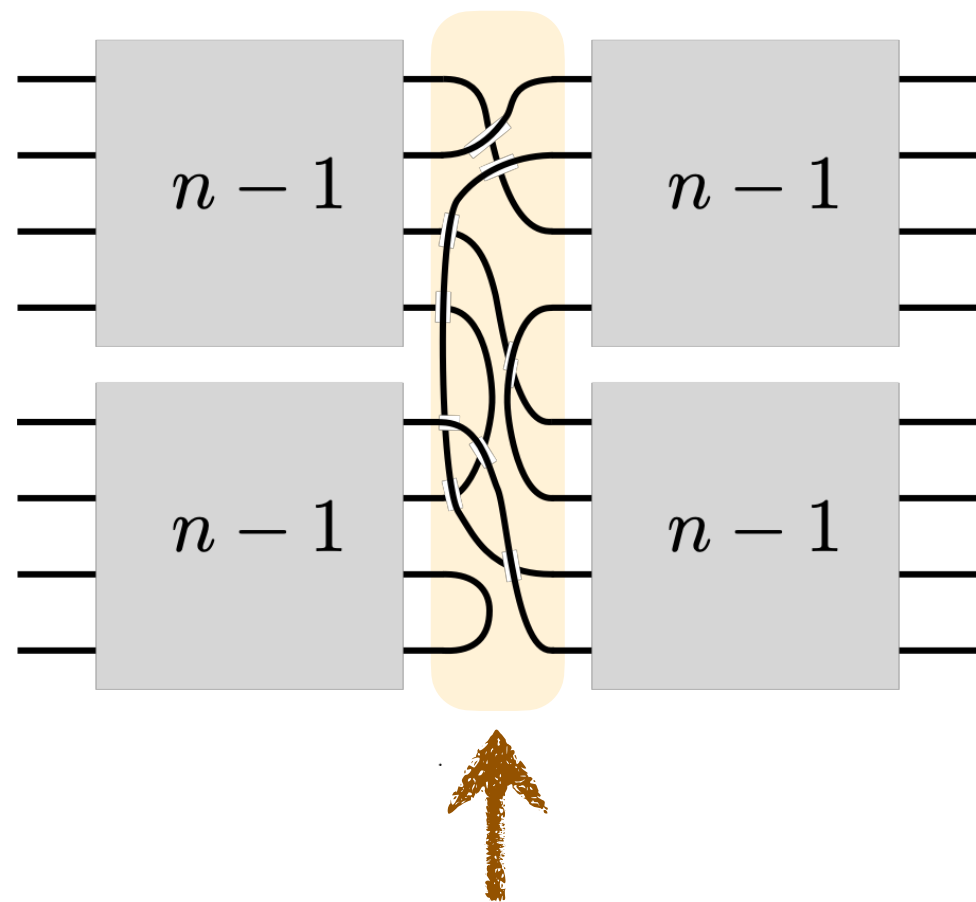
$$A_n = 2^{(d-1)n} A_0$$



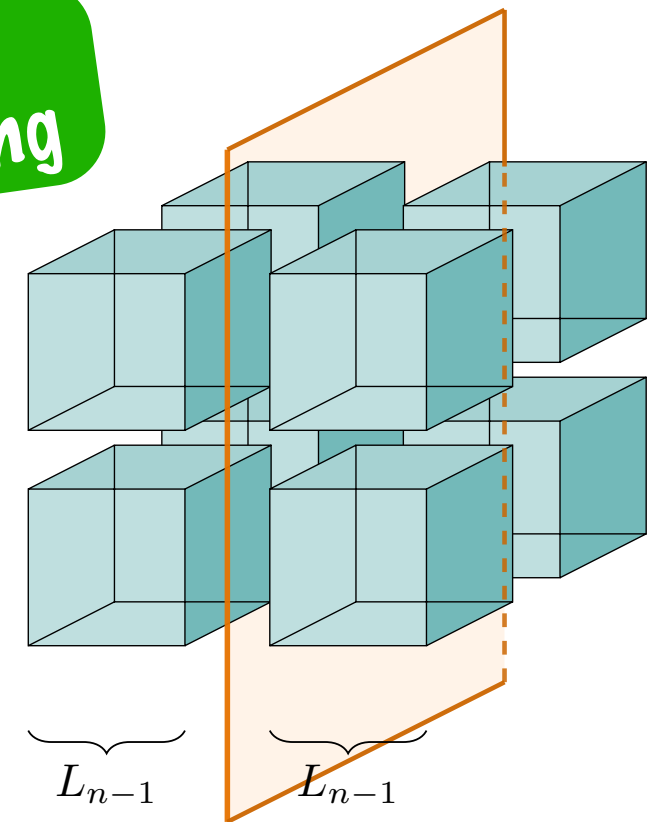
hierarchical construction

2^d independent copies of generation- $(n-1)$ blocks

2^{d-1} copies on the left and right



correct
dimensional scaling



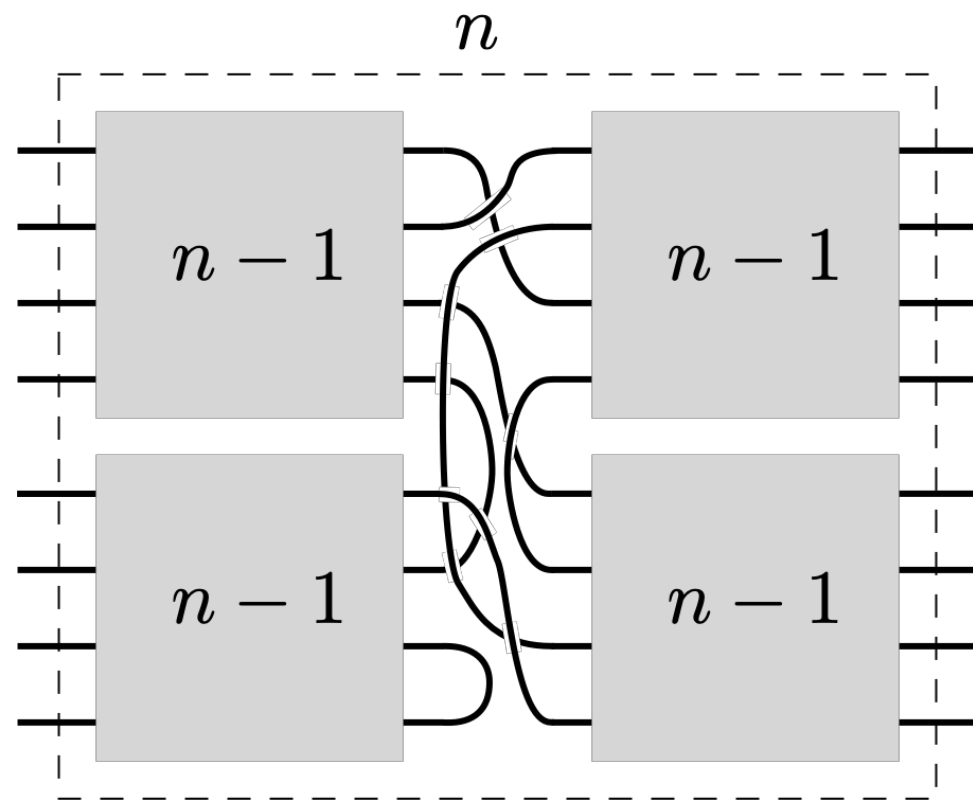
a random matching of $2^d A_{n-1} = 2A_n$
external edges at the interface
(we choose one of the $(2A_n - 1)!!$ perfect
matchings with equal probability)

mean-field-like
long-range matching

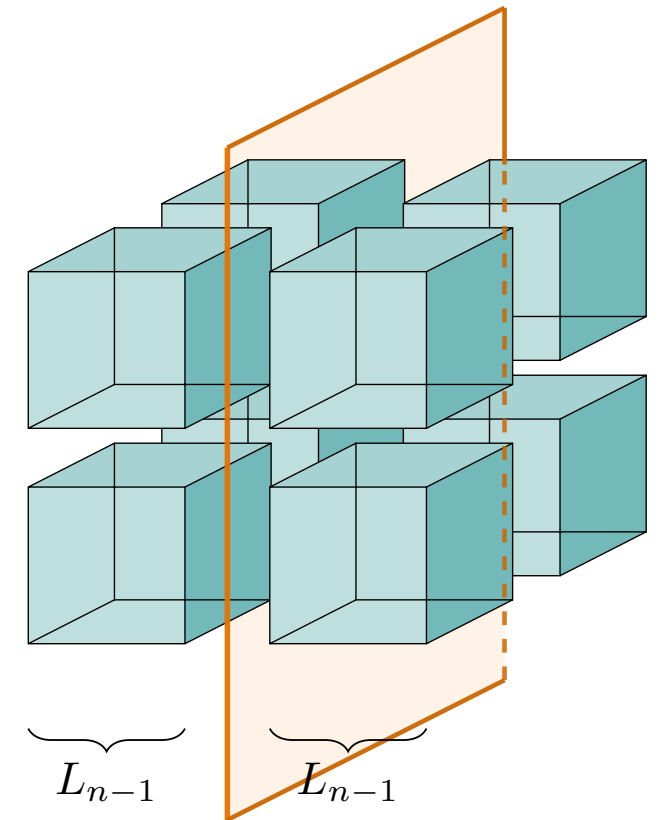
hierarchical construction

2^d independent copies of generation- $(n-1)$ blocks

2^{d-1} copies on the left and right



generation- n block



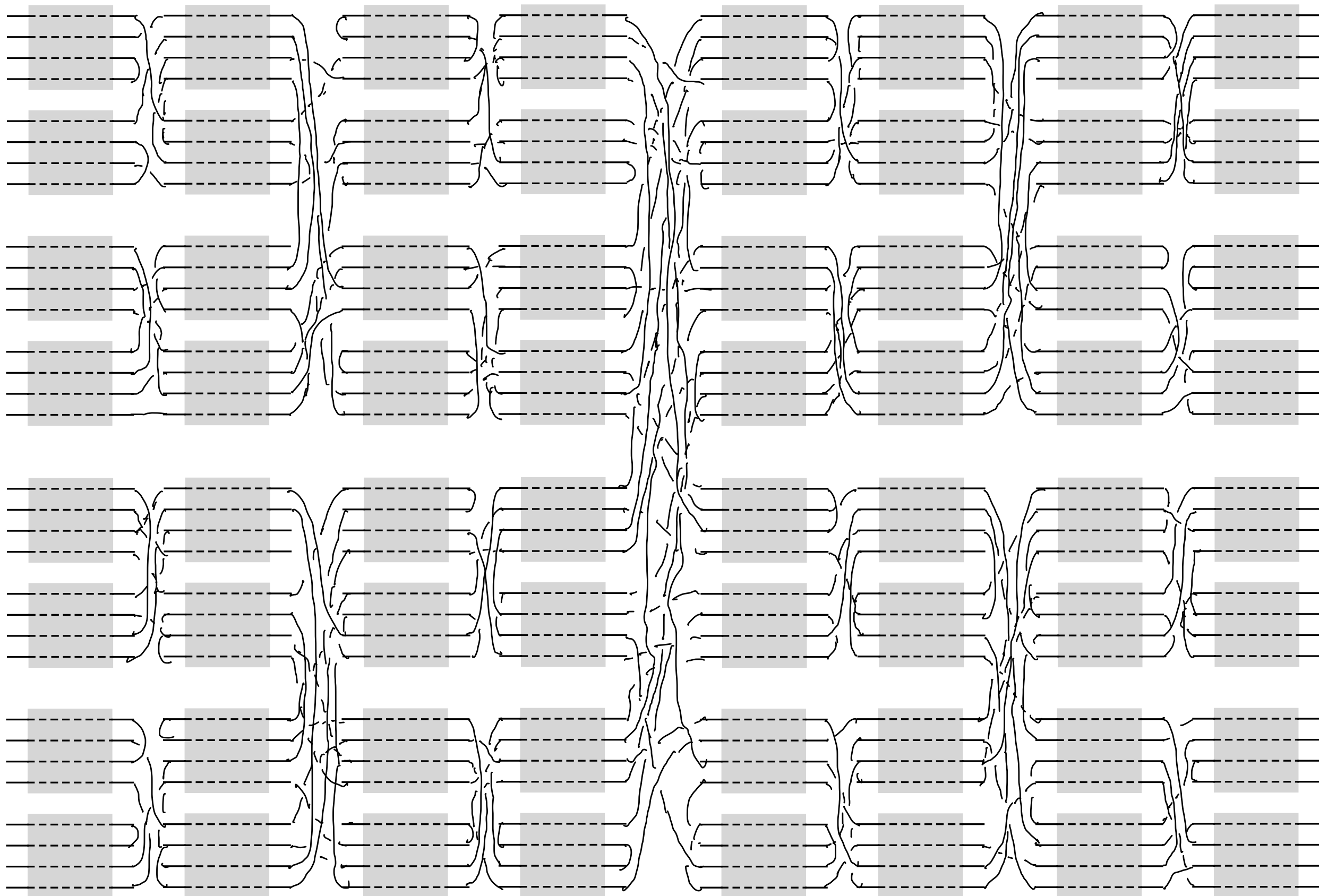
$2^{d-1} A_{n-1} = A_n$ external edges
on the left and right

horizontal length $2L_{n-1} = L_n$

$$A_n = 2^{(d-1)n} A_0 \quad L_n = 2^n$$

hierarchical construction

a generation-3 block ($d = 2, A_0 = 4$)



exact recursion relation

$P_n(\ell)$ probability that there are exactly ℓ crossings in generation- n block ($n = 0, 1, \dots$)

$$P_0(\ell) = \delta_{\ell, A_0}$$

$$P_n(\ell) = \sum_{\ell_1, \dots, \ell_{2^d}} K(\ell \mid \ell_L, \ell_R) \prod_{j=1}^{2^d} P_{n-1}(\ell_j)$$

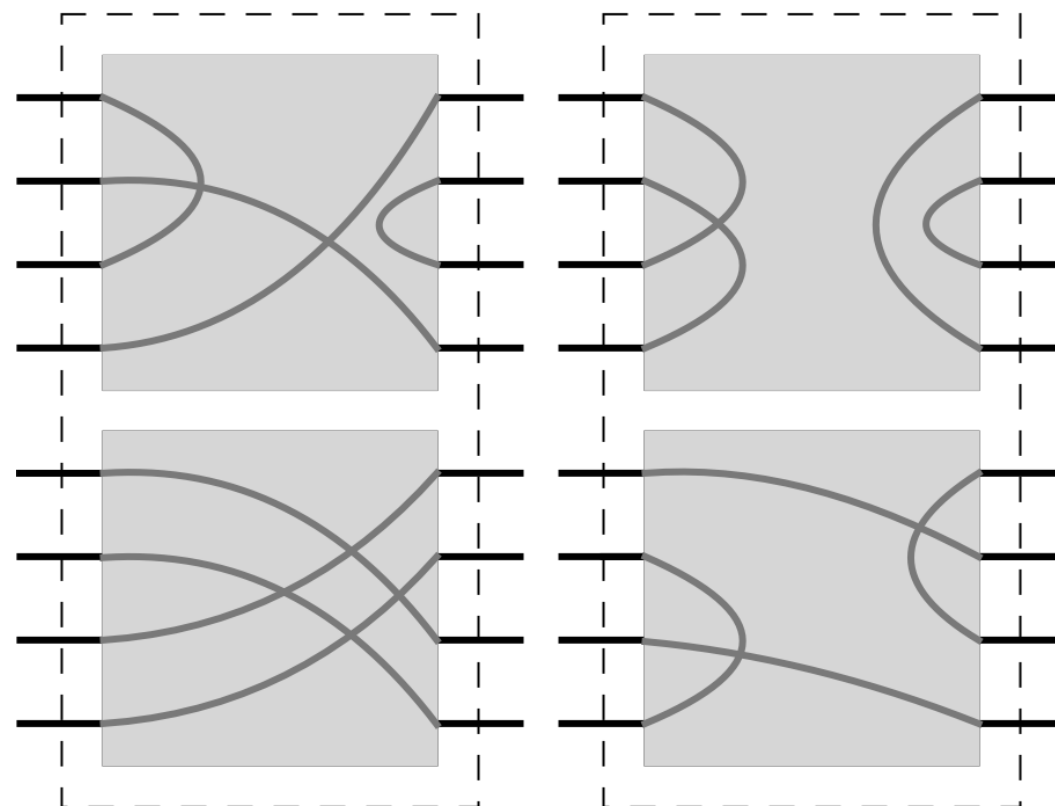
over nonnegative even $\ell_1, \dots, \ell_{2^d}$

$$\ell_L = \sum_{j=1}^{2^{d-1}} \ell_j$$

$$\ell_R = \sum_{j=2^{d-1}+1}^{2^d} \ell_j$$

the number of
crossings on
the left

$$\ell_L = 6$$



the number of
crossings on
the right

$$\ell_R = 2$$

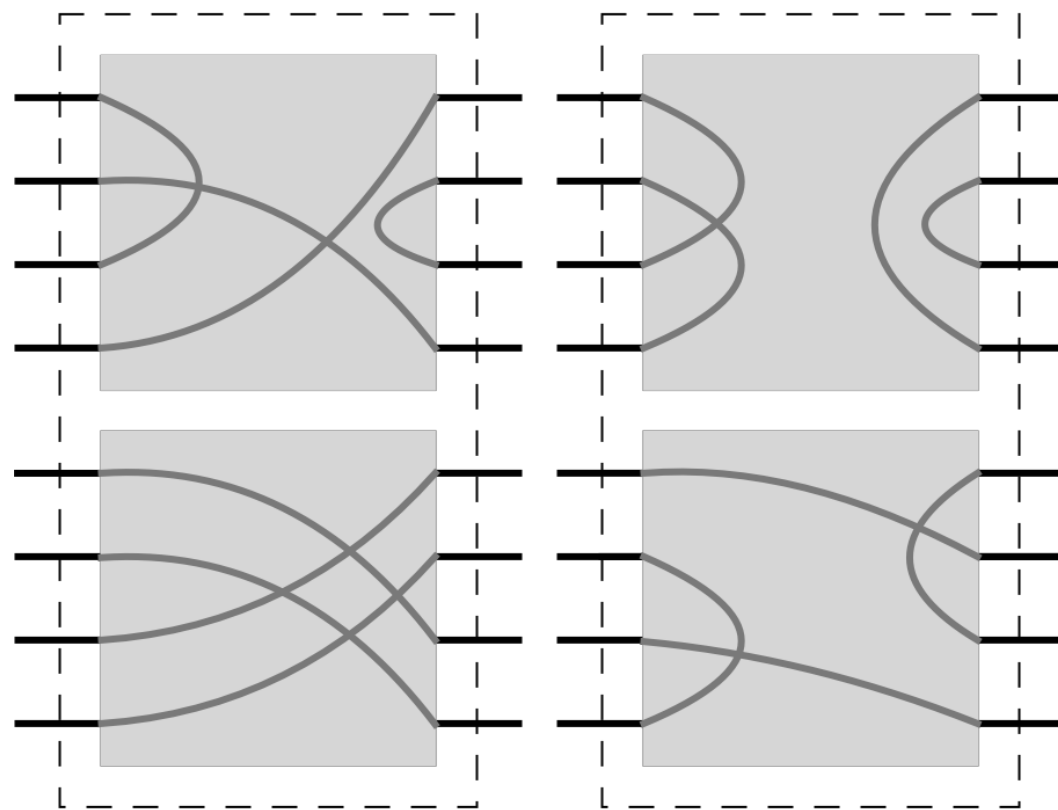
exact recursion relation

$P_n(\ell)$ probability that there are exactly ℓ crossings in generation- n block ($n = 0, 1, \dots$)

$$P_0(\ell) = \delta_{\ell, A_0}$$

$$P_n(\ell) = \sum_{\ell_1, \dots, \ell_{2d}} K(\ell \mid \ell_L, \ell_R) \prod_{j=1}^{2^d} P_{n-1}(\ell_j)$$

ℓ_L crossings
on the left



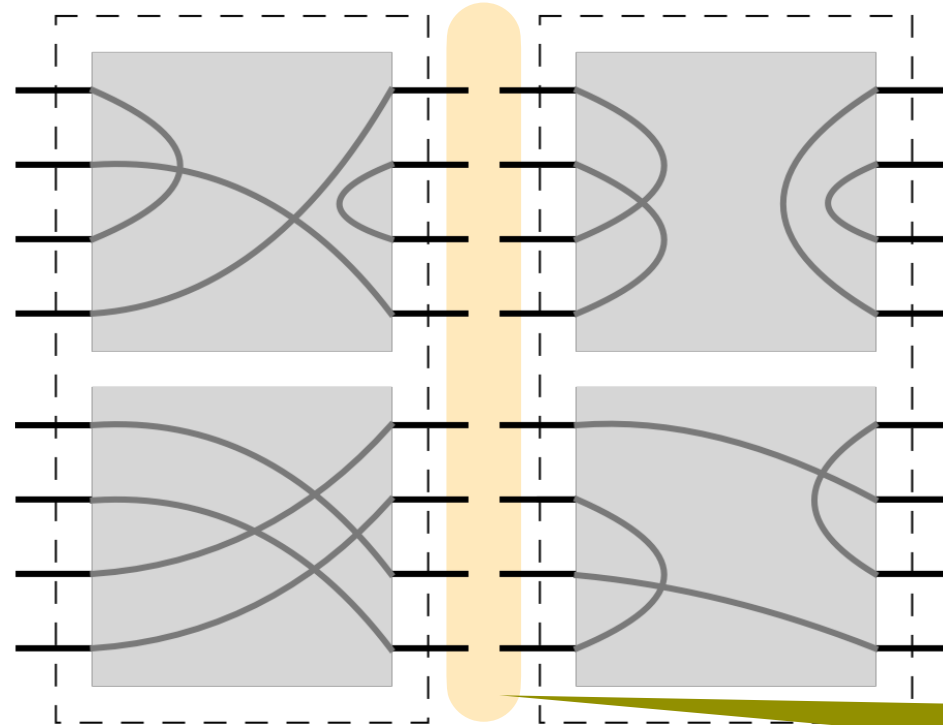
ℓ_R crossings
on the right

$K(\ell \mid \ell_L, \ell_R)$ conditional probability that there are exactly ℓ crossings between the left-most and right-most sides

recursion kernel

$K(\ell \mid \ell_L, \ell_R)$ conditional probability that there are exactly ℓ crossings between the left-most and right-most sides

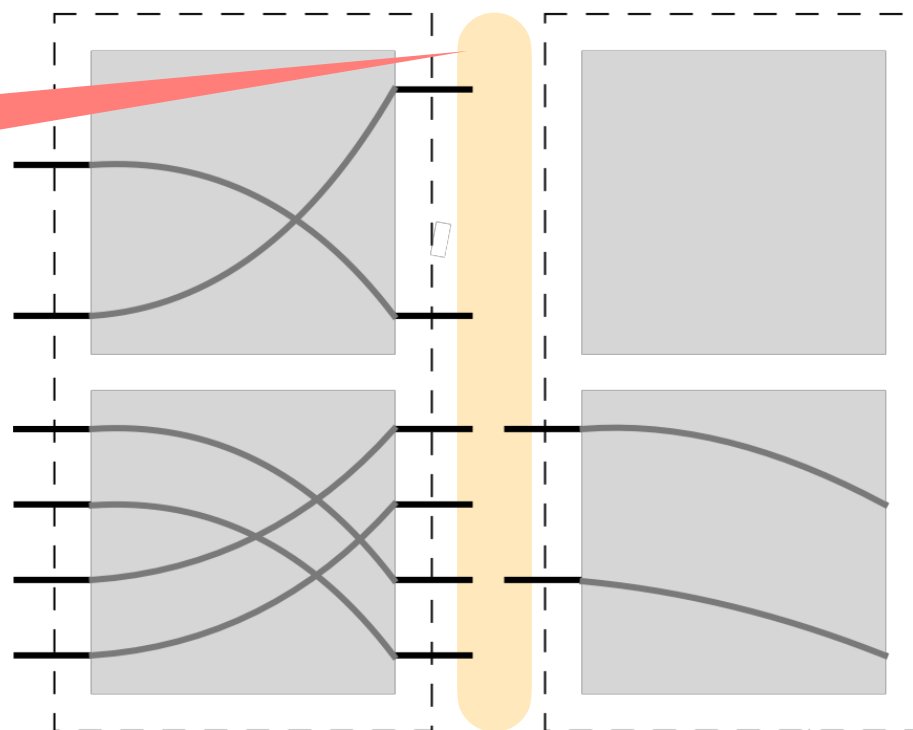
ℓ_L crossings
on the left



ℓ_R crossings
on the right

it suffices to consider

random perfect
matching of only
 $\ell_L + \ell_R$ edges

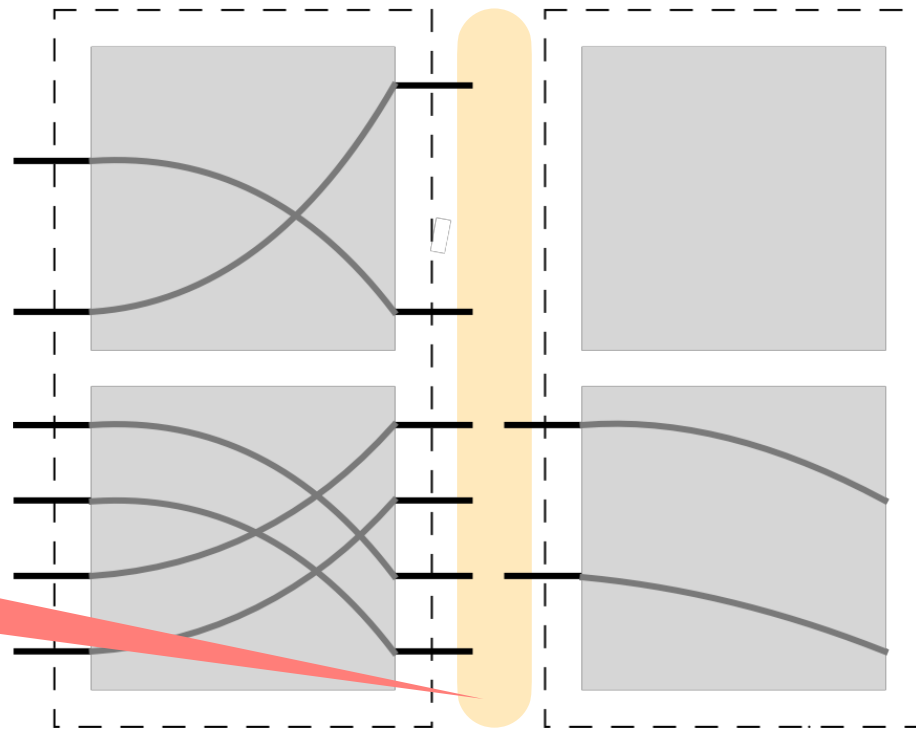


random perfect
matching of all $2A_n$
edges

recursion kernel

$K(\ell \mid \ell_L, \ell_R)$ conditional probability that there are exactly ℓ crossings between the left-most and right-most sides

ℓ_L crossings
on the left



ℓ_R crossings
on the right

random perfect
matching of only
 $\ell_L + \ell_R$ edges

ℓ even integer
 $0 \leq \ell \leq \min\{\ell_L, \ell_R\}$

the number of ways to
form U-turns

the number of ways to choose
 ℓ edges from the left and right

$$K(\ell \mid \ell_L, \ell_R) = \frac{(\ell_L - \ell - 1)!! (\ell_R - \ell - 1)!!}{(\ell_L + \ell_R - 1)!!} \binom{\ell_L}{\ell} \binom{\ell_R}{\ell} \ell!$$

the total number of matchings
of $\ell_L + \ell_R$ edges

the number of ways to
match those edges

exact recursion relation

$P_n(\ell)$ probability that there are exactly ℓ crossings in generation- n block ($n = 0, 1, \dots$)

$$P_0(\ell) = \delta_{\ell, A_0}$$

$$P_n(\ell) = \sum_{\ell_1, \dots, \ell_{2^d}} K(\ell \mid \ell_L, \ell_R) \prod_{j=1}^{2^d} P_{n-1}(\ell_j)$$

over nonnegative even $\ell_1, \dots, \ell_{2^d}$

$$\ell_L = \sum_{j=1}^{2^{d-1}} \ell_j \qquad \ell_R = \sum_{j=2^{d-1}+1}^{2^d} \ell_j$$

$$K(\ell \mid \ell_L, \ell_R) = \frac{(\ell_L - \ell - 1)!! (\ell_R - \ell - 1)!!}{(\ell_L + \ell_R - 1)!!} \binom{\ell_L}{\ell} \binom{\ell_R}{\ell} \ell!$$

if ℓ is an even integer s.t. $0 \leq \ell \leq \min\{\ell_L, \ell_R\}$

$K(\ell \mid \ell_L, \ell_R) = 0$ otherwise

$$\sum_{\ell} K(\ell \mid \ell_L, \ell_R) = 1$$

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behavior of the mean conductance

$P_n(\ell)$ probability that there are exactly ℓ crossings in generation- n block ($n = 0, 1, \dots$)

mean conductance $\mu_n = \sum_{\ell=0,2,\dots} \ell P_n(\ell)$

conditional mean

for $\ell_L, \ell_R \gg 1$

$$\tilde{\mu}(\ell_L, \ell_R) = \sum_{\ell} \ell K(\ell | \ell_L, \ell_R) = \frac{\ell_L \ell_R}{\ell_L + \ell_R + 1} \simeq \left(\frac{1}{\ell_L} + \frac{1}{\ell_R} \right)^{-1}$$

since the main contribution in the recursion should come from $\ell_L \simeq \ell_R \simeq 2^{d-1} \mu_{n-1}$, we expect

$$\mu_n \simeq 2^{d-2} \mu_{n-1}$$

normal transport!!

$$\mu_n \simeq 2^{(d-2)n} \mu_0 = 2^{(d-2)n} A_0 = \frac{A_n}{L_n} \quad \begin{aligned} A_n &= 2^{(d-1)n} A_0 \\ L_n &= 2^n \end{aligned}$$

$$P_n(\ell) = \sum_{\ell_1, \dots, \ell_{2^d}} K(\ell | \ell_L, \ell_R) \prod_{j=1}^{2^d} P_{n-1}(\ell_j)$$
$$\ell_L = \sum_{j=1}^{2^{d-1}} \ell_j \quad \ell_R = \sum_{j=2^{d-1}+1}^{2^d} \ell_j$$

main theorem: normal transport

theorem: let $d \geq 3$. for sufficiently large A_0 there exist positive constants C_d, C'_d such that

$$C_d \frac{A_n}{L_n} - 1 \leq \mu_n \leq C'_d \frac{A_n}{L_n}$$

for all n . if $A_0 \gg 1$, one has $C_d \simeq C'_d \simeq 1$ and hence $\mu_n \simeq A_n/L_n$ for all n .

for $d = 3$, $A_0 \geq 2$ suffices (no conditions)
 $C_3 = 1 - 5/(4A_0)$, $C'_3 = 1 + 1/(3A_0)$

normal transport from non-chaotic deterministic motion in a quenched random environment

proposition: let $d = 1$ and assume $A_0 \gg 1$. then $\mu_n \simeq A_0/L_n$ as long as μ_n is larger than $O(1)$

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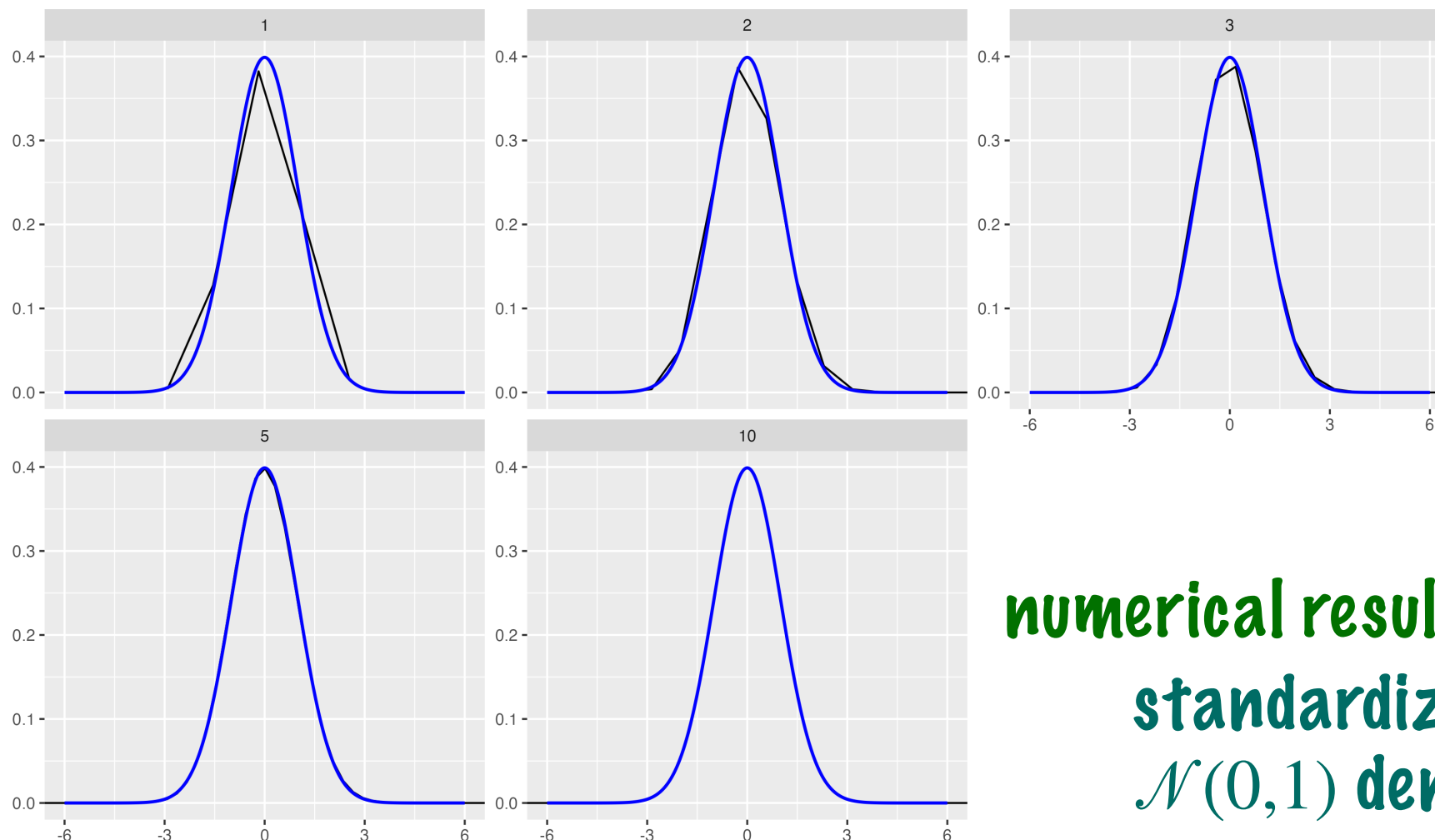
basic ansatz and approximation

$P_n(\ell)$ probability that there are exactly ℓ crossings in generation- n block ($n = 0, 1, \dots$)

mean conductance $\mu_n = \sum_{\ell=0,2,\dots} \ell P_n(\ell)$

sample-to-sample variance $v_n = \sum_{\ell=0,2,\dots} (\ell - \mu_n)^2 P_n(\ell)$

ansatz $P_n(\ell) \propto \exp\left[-\frac{(\ell - \mu_n)^2}{2v_n}\right]$



numerical results for $d = 3$ with $A_0 = 2$
standardized $P_n(\ell)$ vs $\mathcal{N}(0,1)$ density for $n = 1, 2, 3, 5, 10$

basic ansatz and approximation

$P_n(\ell)$ probability that there are exactly ℓ crossings in generation- n block ($n = 0, 1, \dots$)

mean conductance $\mu_n = \sum_{\ell=0,2,\dots} \ell P_n(\ell)$

sample-to-sample variance $v_n = \sum_{\ell=0,2,\dots} (\ell - \mu_n)^2 P_n(\ell)$

ansatz $P_n(\ell) \propto \exp\left[-\frac{(\ell - \mu_n)^2}{2v_n}\right]$

approximation

$$K(\ell \mid \ell_L, \ell_R) = \frac{(\ell_L - \ell - 1)!! (\ell_R - \ell - 1)!!}{(\ell_L + \ell_R - 1)!!} \binom{\ell_L}{\ell} \binom{\ell_R}{\ell} \ell!$$

$$\propto \exp\left[-\frac{\{\ell - \tilde{\mu}(\ell_L, \ell_R)\}^2}{2\tilde{v}(\ell_L, \ell_R)}\right]$$

$$\tilde{\mu}(\ell_L, \ell_R) = \frac{\ell_L \ell_R}{\ell_L + \ell_R + 1}$$

$$\tilde{v}(\ell_L, \ell_R) = \frac{2\ell_L \ell_R (\ell_L - 1)(\ell_R - 1)}{(\ell_L + \ell_R - 3)(\ell_L + \ell_R - 1)^2}$$

recursion relations for μ_n and v_n

ansatz $P_n(\ell) \propto \exp\left[-\frac{(\ell - \mu_n)^2}{2v_n}\right]$

approximation $K(\ell | \ell_L, \ell_R) \propto \exp\left[-\frac{\{\ell - \tilde{\mu}(\ell_L, \ell_R)\}^2}{2\tilde{v}(\ell_L, \ell_R)}\right]$

$$\tilde{\mu}(\ell_L, \ell_R) = \frac{\ell_L \ell_R}{\ell_L + \ell_R + 1} \quad \tilde{v}(\ell_L, \ell_R) = \frac{2\ell_L \ell_R (\ell_L - 1)(\ell_R - 1)}{(\ell_L + \ell_R - 3)(\ell_L + \ell_R - 1)^2}$$

recursion

$$P_n(\ell) = \sum_{\ell_1, \dots, \ell_{2^d}} K(\ell | \ell_L, \ell_R) \prod_{j=1}^{2^d} P_{n-1}(\ell_j)$$
$$\ell_L = \sum_{j=1}^{2^{d-1}} \ell_j \quad \ell_R = \sum_{j=2^{d-1}+1}^{2^d} \ell_j$$

for $d \geq 3$, the leading terms are

$$\mu_n \simeq 2^{d-2} \mu_{n-1}$$

$$v_n \simeq 2^{d-3} \mu_{n-1} + 2^{d-4} v_{n-1}$$

normal transport

for $d \geq 3$, the leading terms are

$$\mu_n \simeq 2^{d-2} \mu_{n-1}$$

$$v_n \simeq 2^{d-3} \mu_{n-1} + 2^{d-4} v_{n-1}$$

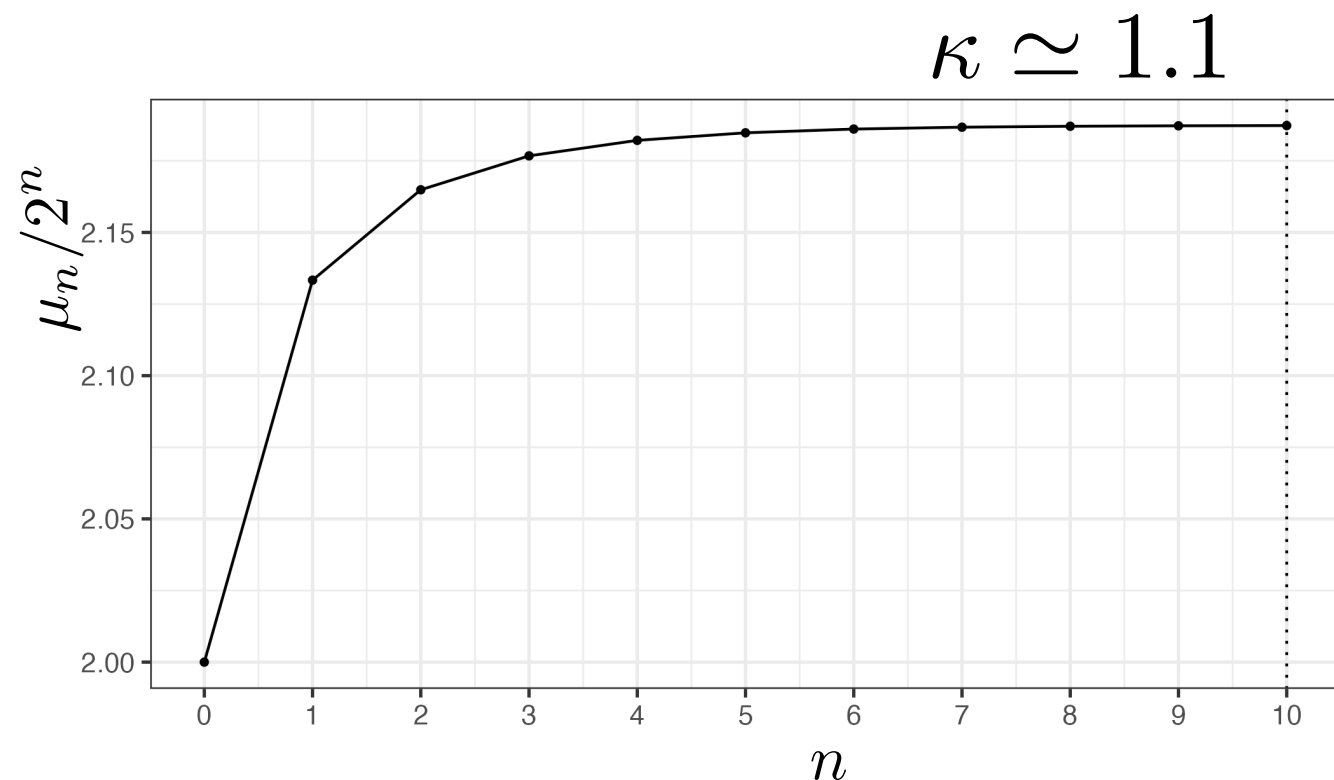
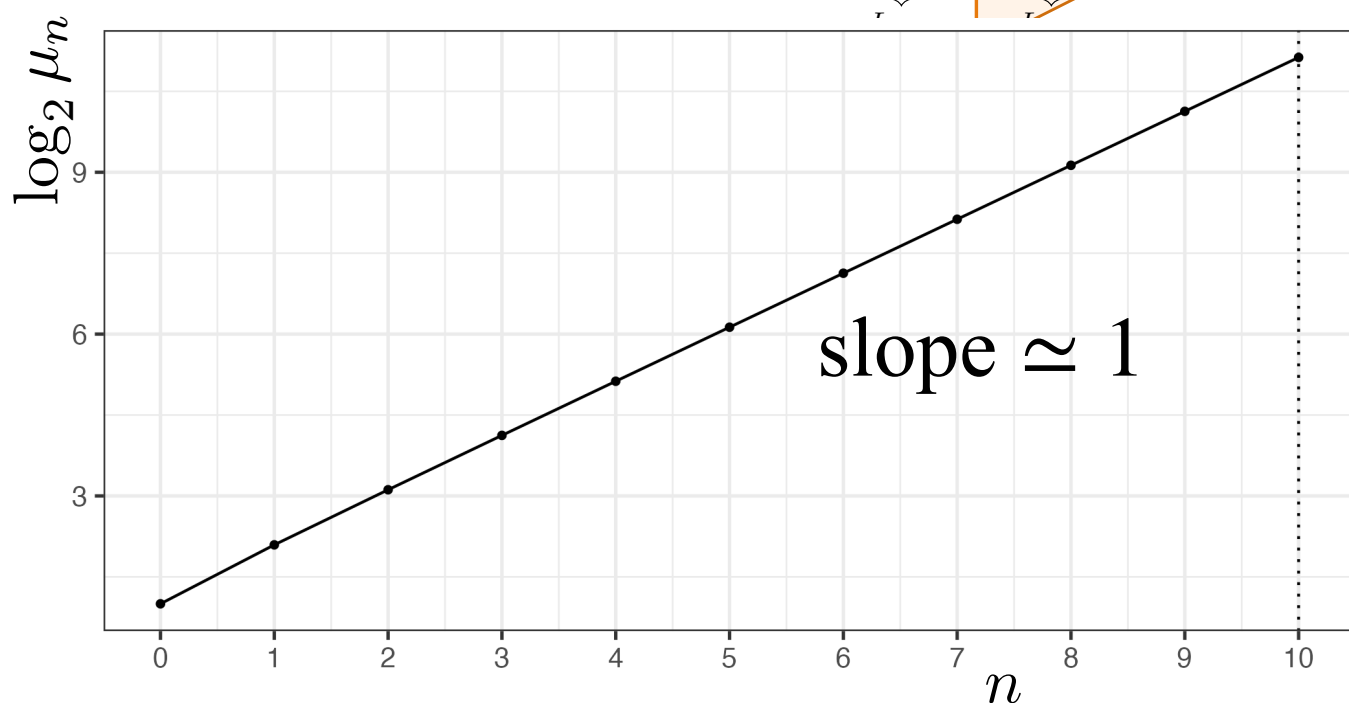
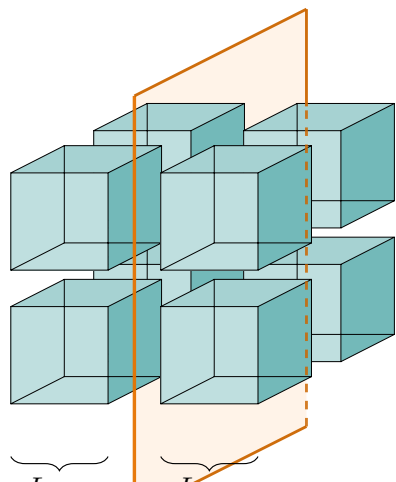
$$\mu_n \simeq (\text{const}) 2^{(d-2)n} = \kappa \frac{A_n}{L_n} \text{ as } n \uparrow \infty$$

conductivity

normal transport

proven rigorously!

numerical results
for $d = 3$ with $A_0 = 2$



universal “2/3 law”

for $d \geq 3$, the leading terms are

$$\mu_n \simeq 2^{d-2} \mu_{n-1}$$

$$v_n \simeq 2^{d-3} \mu_{n-1} + 2^{d-4} v_{n-1}$$

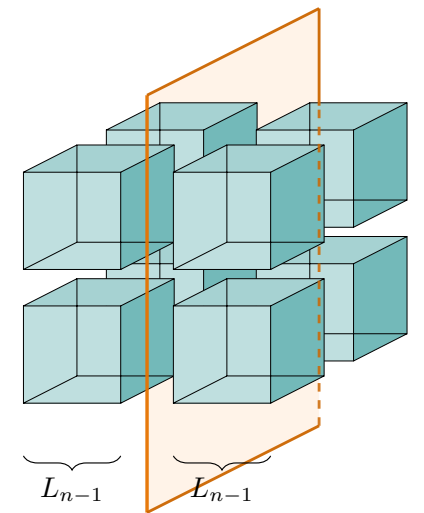
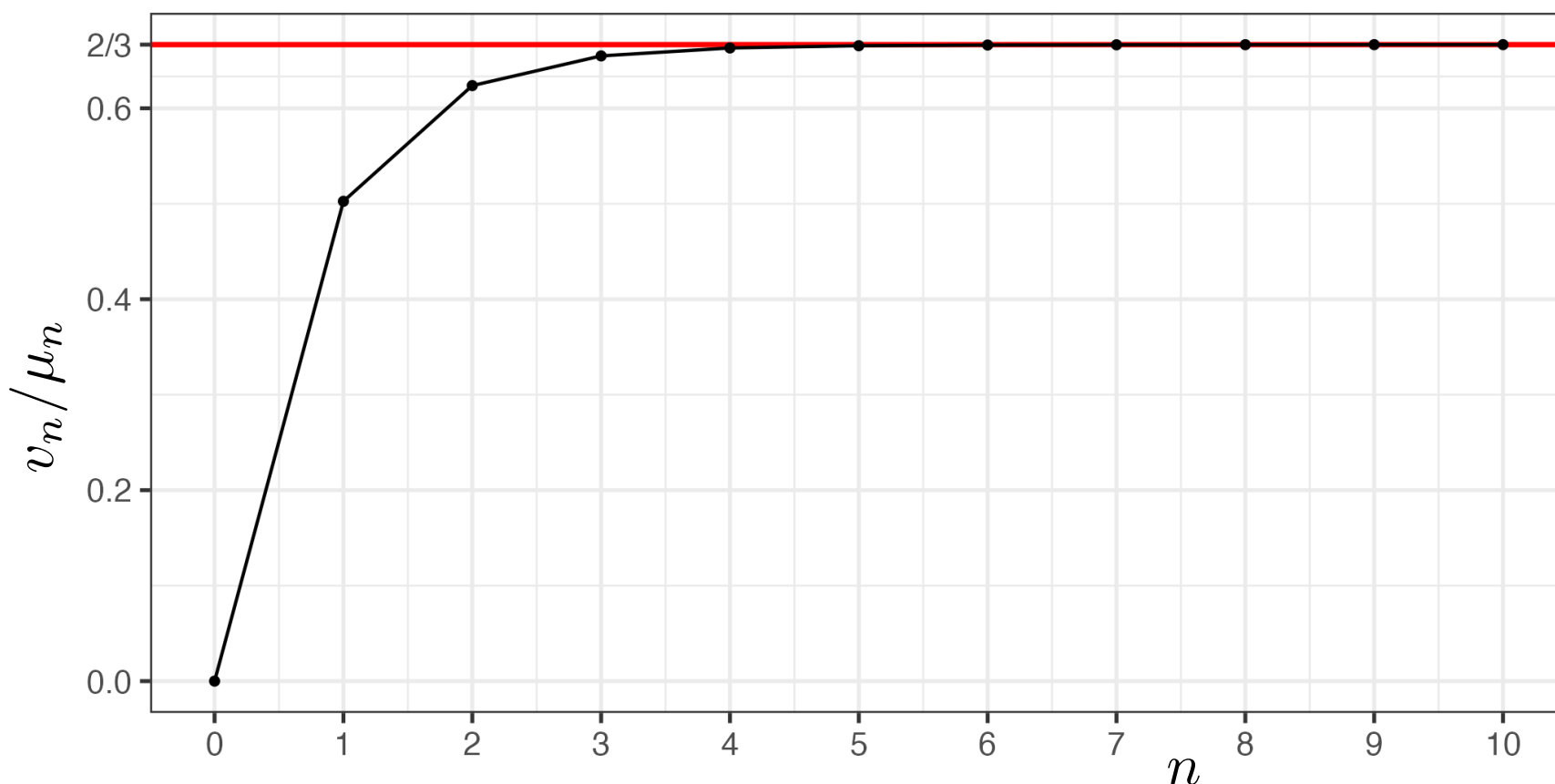
fluctuation from
random matching

fluctuation from
generation $n-1$

$$\frac{v_n}{\mu_n} \simeq \frac{1}{2} + \frac{1}{4} \frac{v_{n-1}}{\mu_{n-1}}$$

d -independent recursion relation

$\frac{v_n}{\mu_n} \rightarrow \frac{2}{3}$ rapidly as $n \uparrow \infty$ in any dimension $d \geq 3$



numerical results
for $d = 3$ with $A_0 = 2$

marginal dimension $d = 2$

for $d = 2$, we include higher order contributions

$$\mu_n \simeq \mu_{n-1} + \frac{1}{4} - \frac{v_{n-1}}{4\mu_{n-1}} \quad v_n \simeq \frac{\mu_{n-1}}{2} + \frac{v_{n-1}}{4}$$

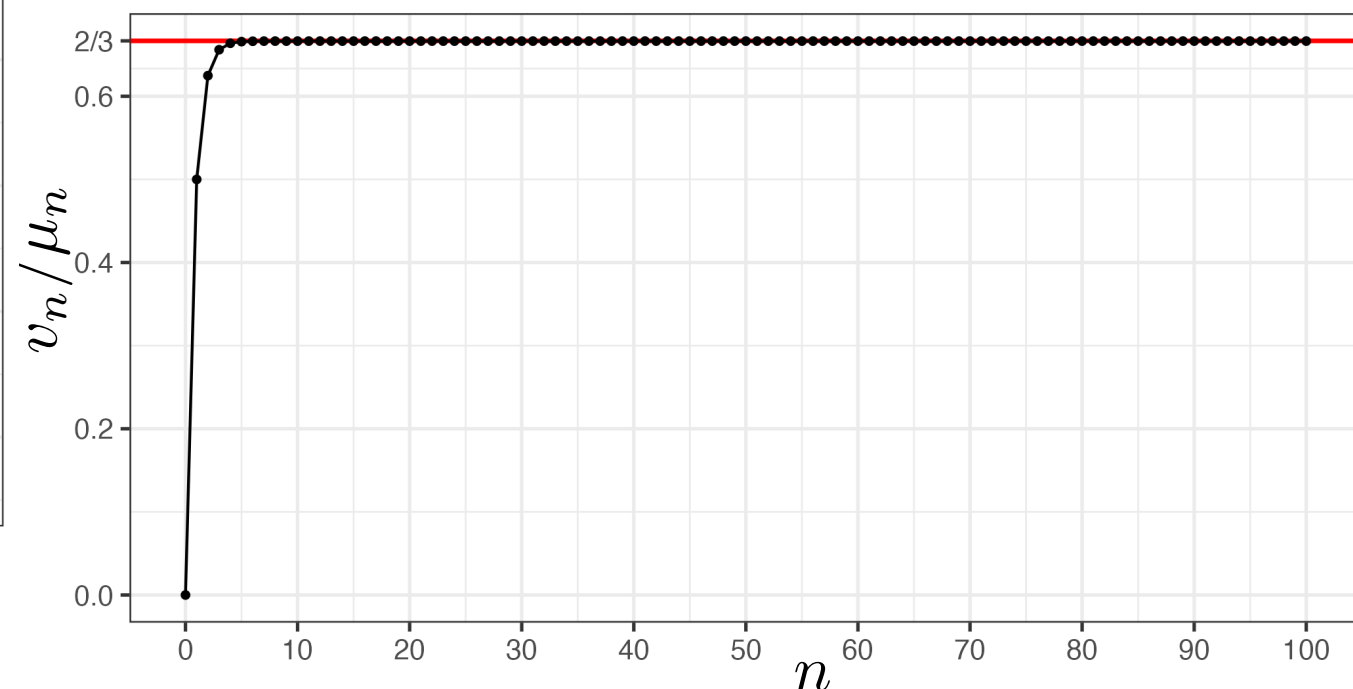
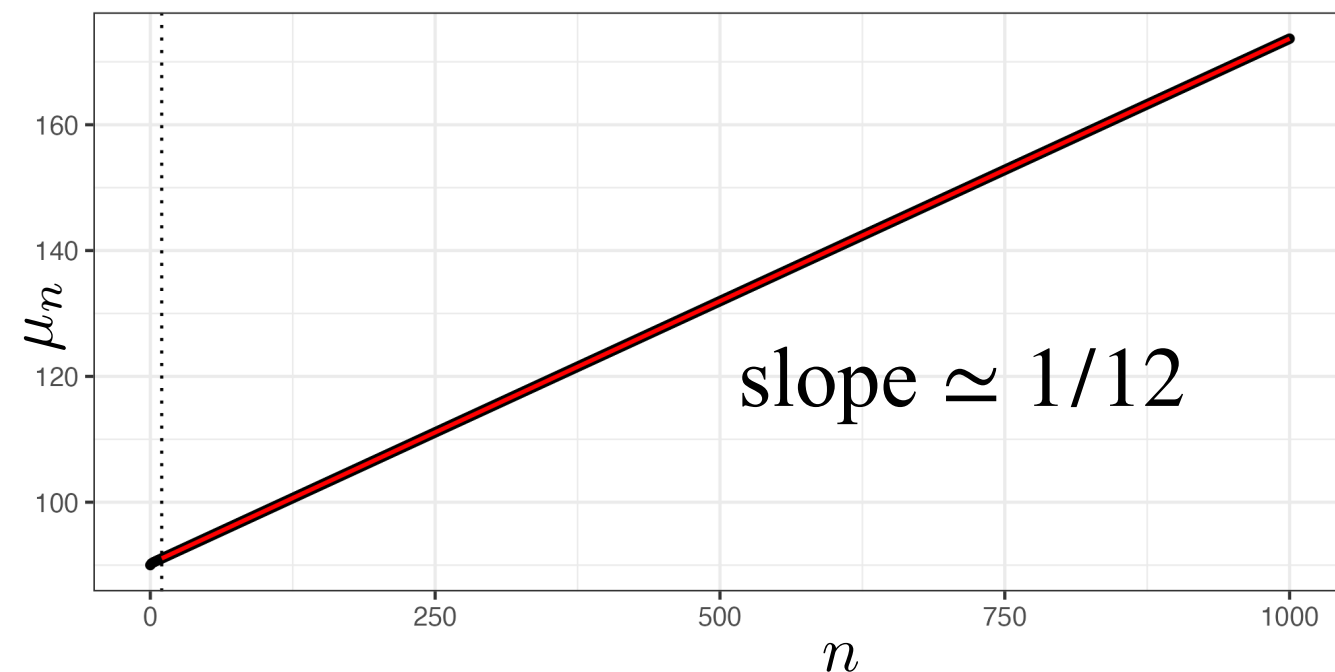
as $n \uparrow \infty$ $\mu_n \simeq \frac{n}{12} = \frac{\log L_n}{12 \log 2}$

weakly anomalous transport
with a logarithmic correction

$v_n \simeq \frac{n}{18}$ and hence $\frac{v_n}{\mu_n} \rightarrow \frac{2}{3}$

2/3 law!!

numerical results for $d = 2$ with $A_0 = 90$



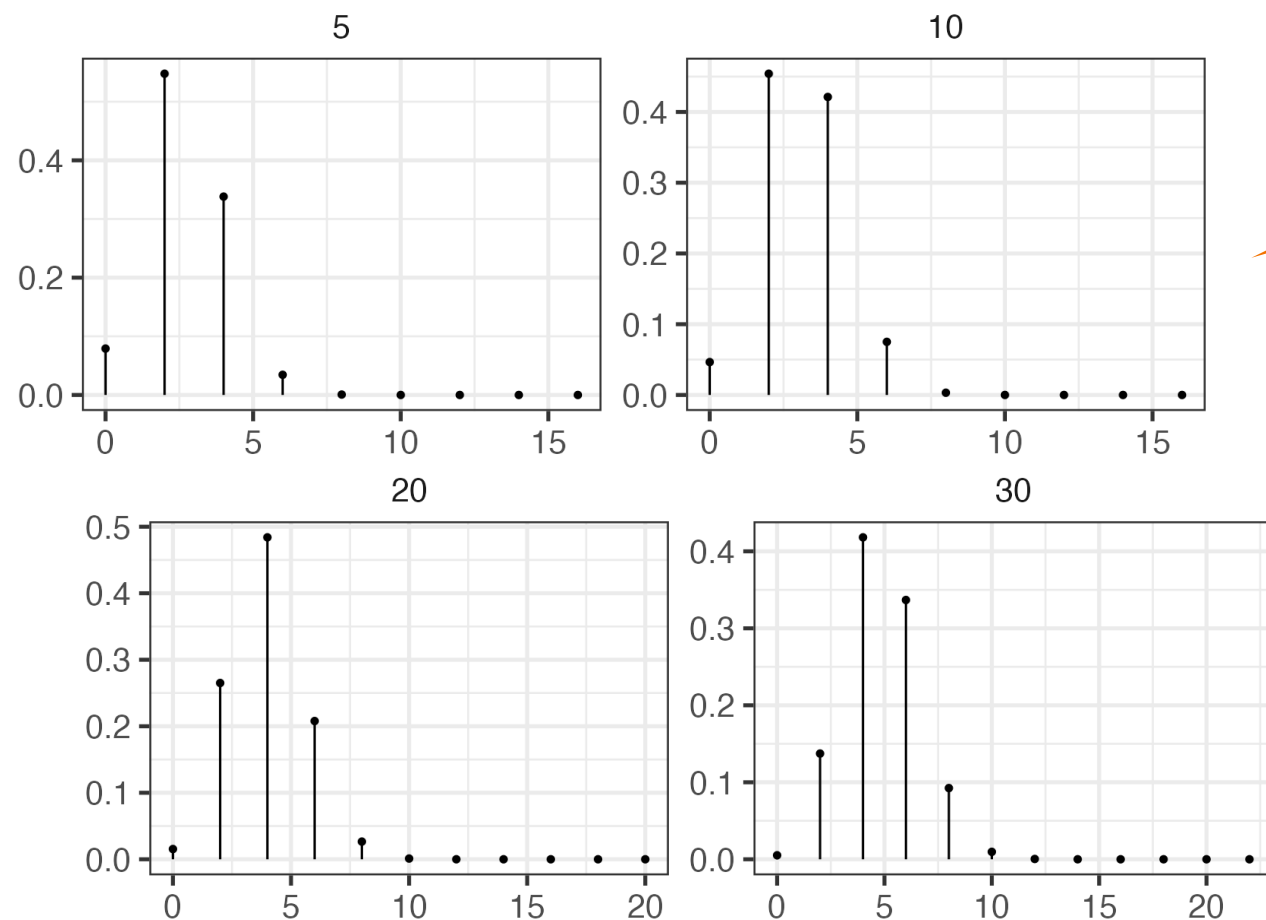
robustness of the “2/3 law”

$$\frac{v_n}{\mu_n} \rightarrow \frac{2}{3} \text{ rapidly whenever } \mu_n \uparrow \infty$$

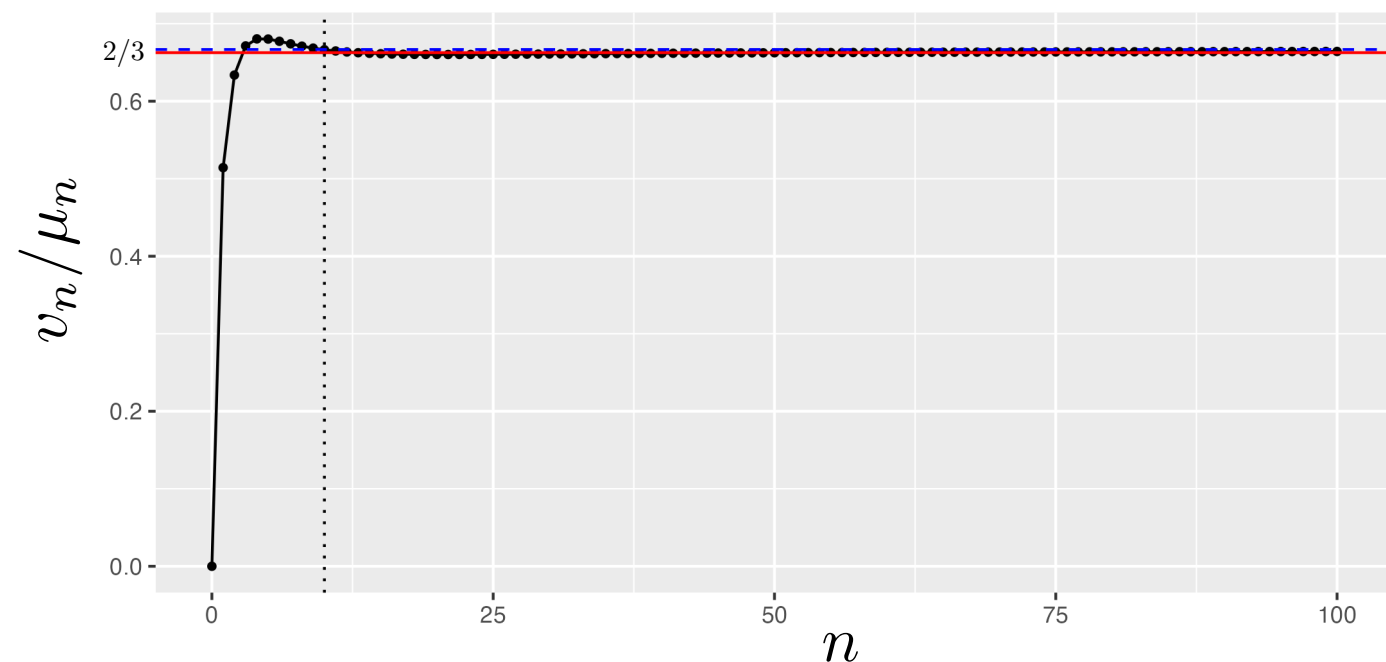
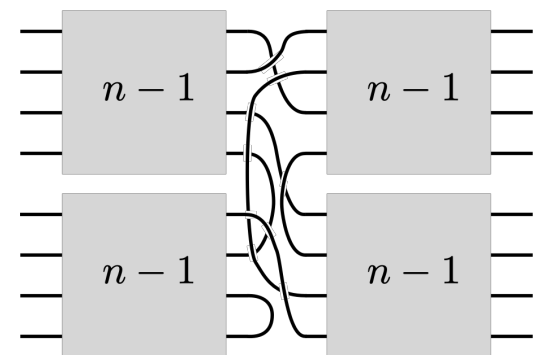
“2/3 law” may be much more robust than the Gaussianity

numerical results for $d = 2$ with $A_0 = 2$

$P_n(\ell)$ vs ℓ for $n = 5, 10, 20, 30$



very far from Gauss!!!



Lorentz mirror model

Hierarchical model

definition and recursion relation

main theorem on normal transport

Gaussian closure and the “2/3 law”

Lorentz mirror model in $d = 3$: numerics

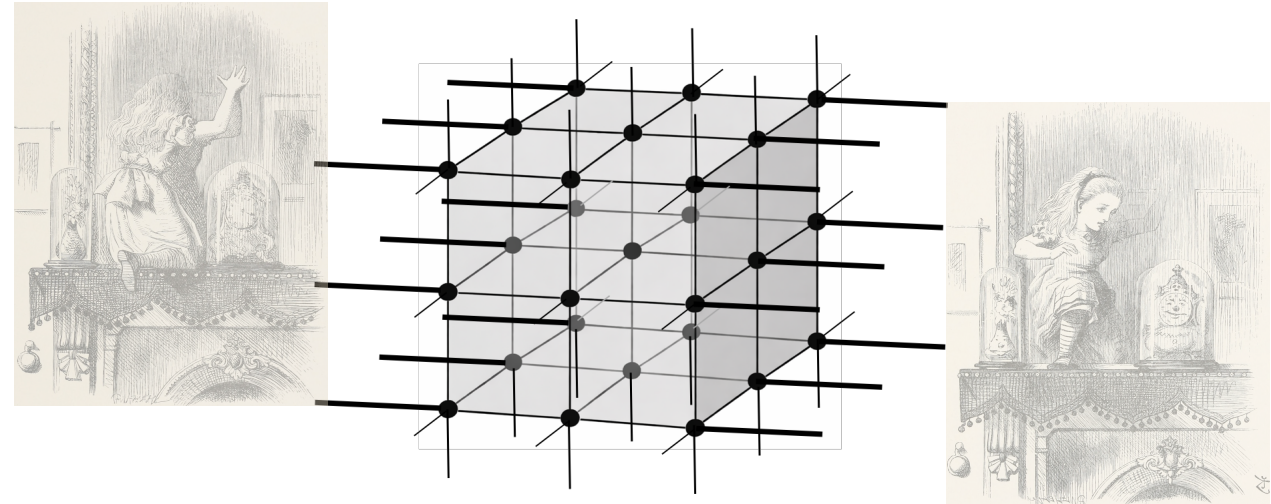
summary

appendix: brief ideas of the proof

Monte Carlo simulation of the original Lorentz mirror model on the cubic lattice

the $L \times L \times L$ cubic lattice
 $L = 2^n$ with $n = 1, 2, \dots, 7$

choose one of the $5!! = 15$
pairings at each vertex



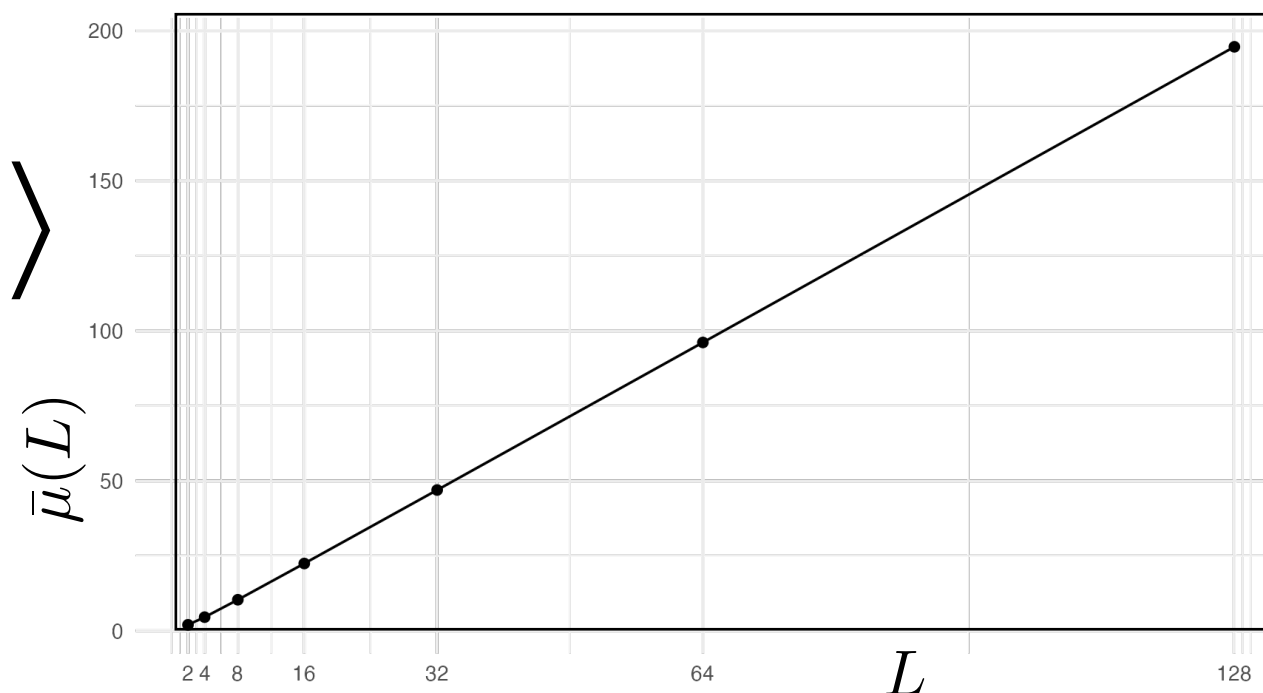
generate 6.4×10^5 independent environments
for each realization of the environment, faithfully
count the number of crossings $\mathcal{C}$, i.e., the conductance

mean conductance $\bar{\mu}(L) = \langle \mathcal{C} \rangle$

variance $\bar{v}(L) = \langle (\mathcal{C} - \bar{\mu}(L))^2 \rangle$

we recover normal transport

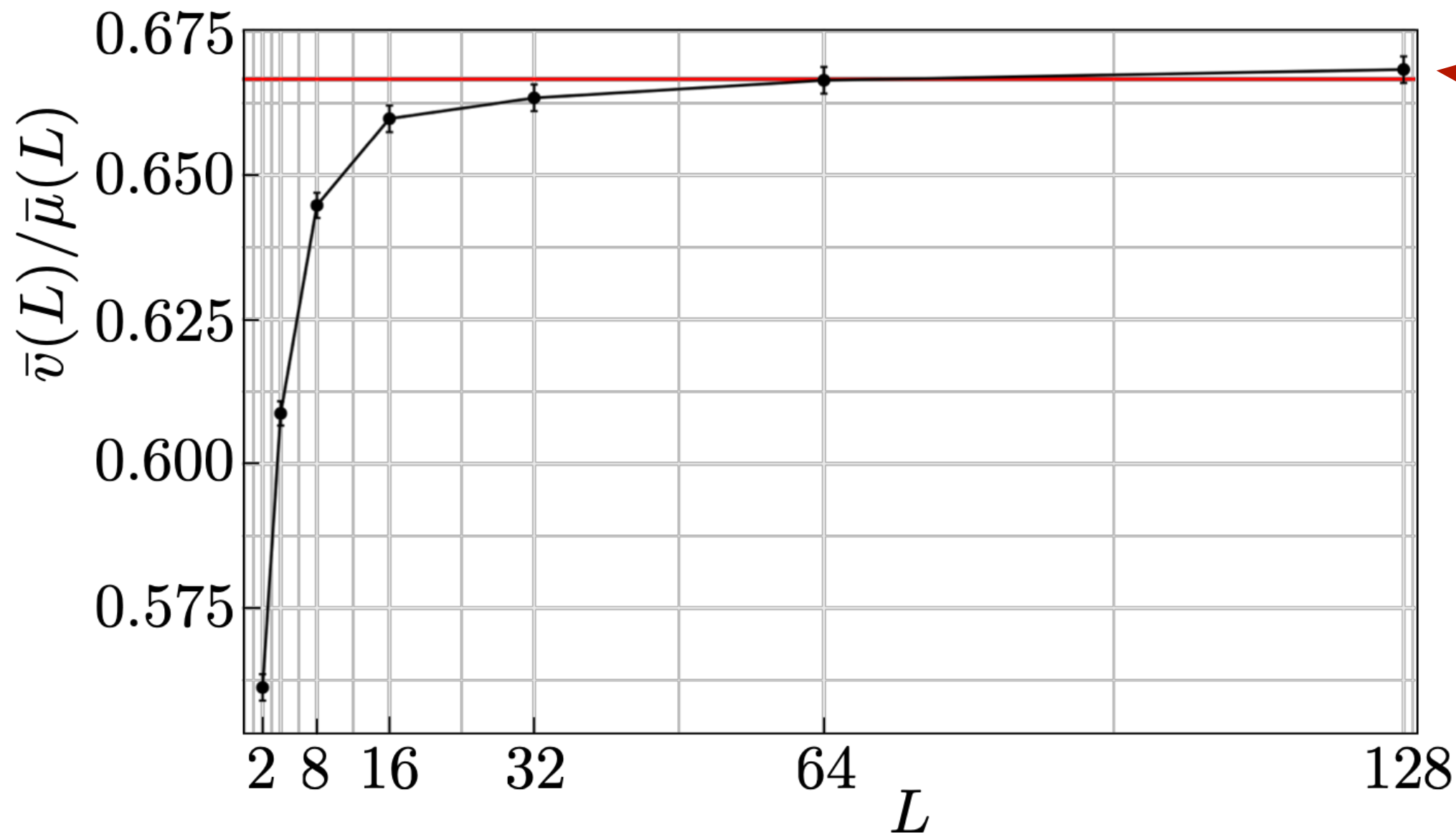
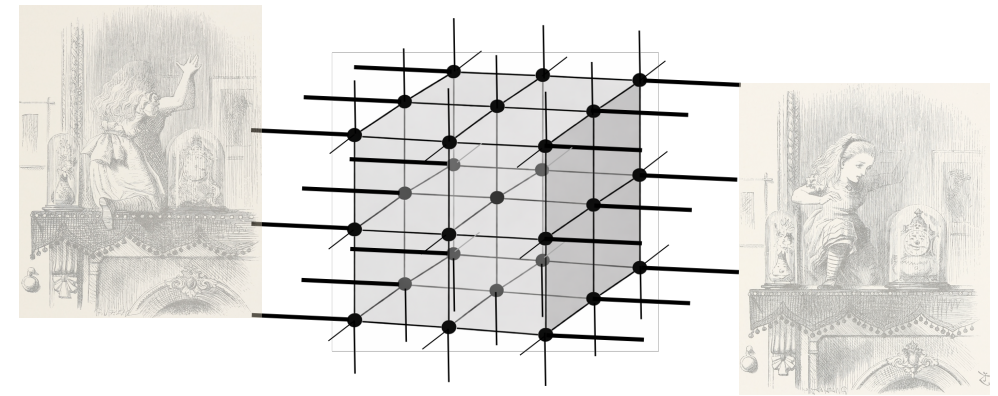
Chiffaudel, Lefevre 2016



variance-to-mean ratio

mean conductance $\bar{\mu}(L) = \langle \mathcal{C} \rangle$

variance $\bar{v}(L) = \langle (\mathcal{C} - \bar{\mu}(L))^2 \rangle$



$\frac{2}{3}!!$

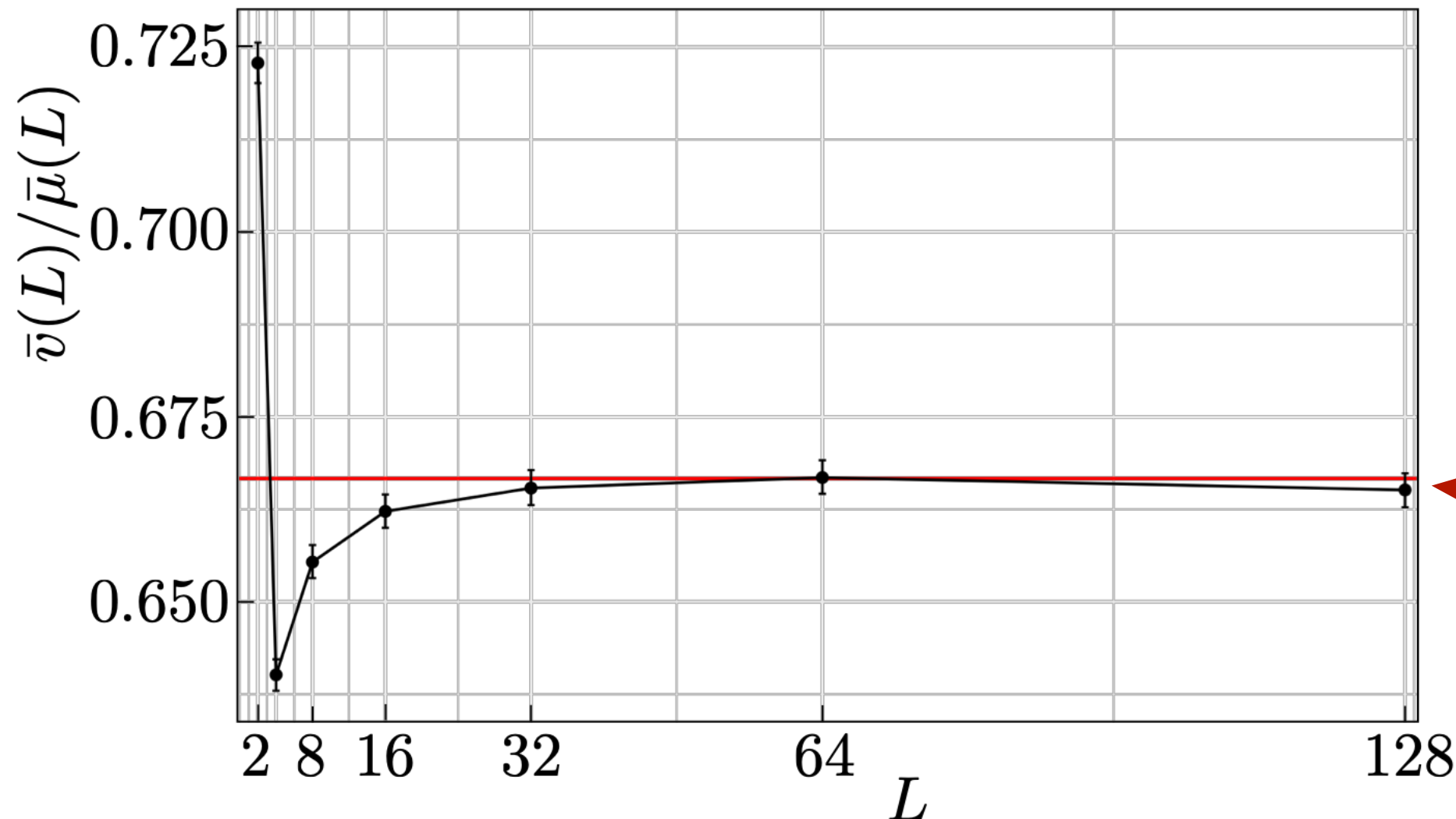
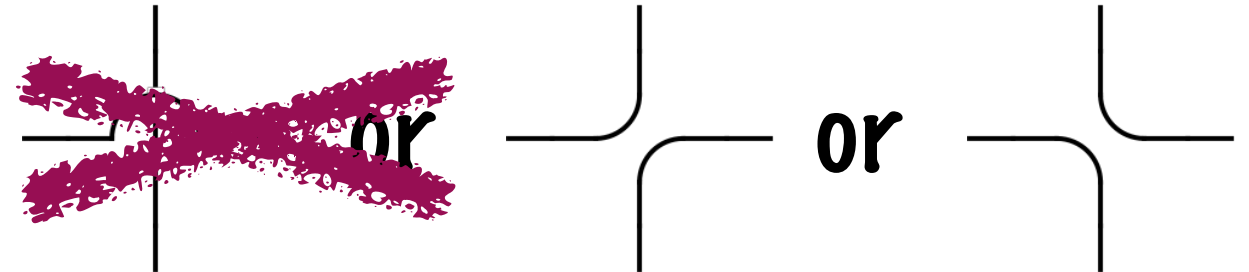
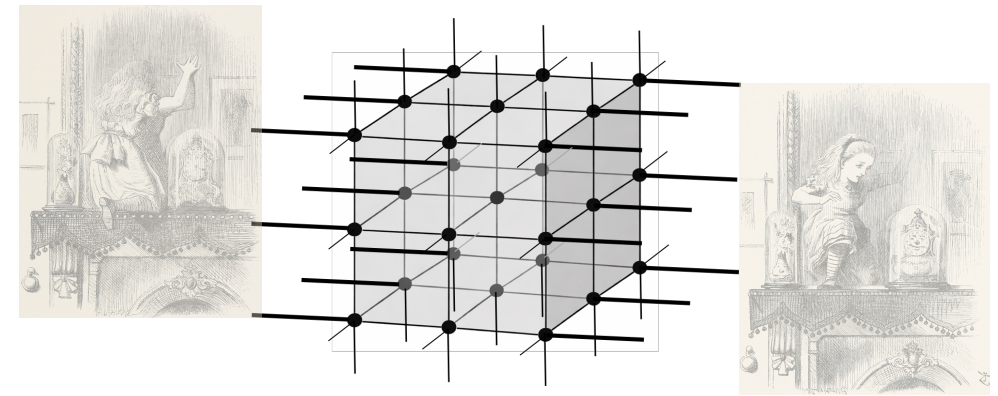
Monte Carlo with 6.4×10^5 samples
error bars indicate 95% confidence intervals

variance-to-mean ratio

“orthogonal rule”

particles never go straight

choose one of the 8 orthogonal pairings at each vertex



still $\frac{2}{3}$!!!!

Monte Carlo with 6.4×10^5 samples
error bars indicate 95% confidence intervals

the universal “2/3 law”

$$\frac{(\text{sample-to-sample variance of conductance})}{(\text{mean conductance})} \rightarrow \frac{2}{3}$$

observed theoretically (but not proved!) in the hierarchical Lorentz mirror model in $d \geq 2$, and numerically in the original Lorentz mirror model in $d = 3$

conjecture: v/μ (as $\mu \uparrow \infty$) is a universal amplitude ratio, and the value $2/3$ signifies a universality class of systems (including ours) exhibiting normal transport induced by random matching of conserved current

what models belong to this universality class?
any models in the continuum as in the original Lorentz gas?

Lorentz mirror model

Hierarchical model

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main theorem on normal transport

Gaussian closure and the “2/3 law”

Lorentz mirror model in $d = 3$: numerics

summary

appendix: brief ideas of the proof

summary

- ✓ the Lorentz mirror model (both original and hierarchical) provides a clean setting to study transport generated by a quenched random environment
- ✓ in the hierarchical version, we proved normal transport in $d \geq 3$ and showed (but didn't prove) weakly anomalous transport with log-correction in $d = 2$
- ✓ in the original model, normal transport has been expected for $d \geq 3$. we conjecture weakly anomalous transport with log-correction in $d = 2$

no numerics yet

- ✓ we discovered the “2/3 law” for the variance-to-mean ratio of conductance, and propose that it is a signature of a universality class of normal transport induced by random current matching

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main theorem: normal transport

theorem: let $d \geq 3$. for sufficiently large A_0 there exist positive constants C_d, C'_d such that

$$C_d \frac{A_n}{L_n} - 1 \leq \mu_n \leq C'_d \frac{A_n}{L_n}$$

for all n . if $A_0 \gg 1$, one has $C_d \simeq C'_d \simeq 1$ and hence $\mu_n \simeq A_n/L_n$ for all n .

for $d = 3, A_0 \geq 2$ suffices

$$C_3 = 1 - 5/(4A_0), C'_3 = 1 + 1/(3A_0)$$

normal transport from non-chaotic deterministic motion in a quenched random environment

proposition: let $d = 1$ and assume $A_0 \gg 1$. then $\mu_n \simeq A_0/L_n$ as long as μ_n is larger than $O(1)$

upper bound

mean conductance $\mu_n = \sum_{\ell} \ell P_n(\ell)$

recursion $P_n(\ell) = \sum_{\ell_1, \dots, \ell_{2^d}} K(\ell | \ell_L, \ell_R) \prod_{j=1}^{2^d} P_{n-1}(\ell_j)$

$$\mu_n = \sum_{\ell_1, \dots, \ell_{2^d}} \boxed{\sum_{\ell} \ell K(\ell | \ell_L, \ell_R)} \prod_{j=1}^{2^d} P_{n-1}(\ell_j)$$

$$\tilde{\mu}(\ell_L, \ell_R) = \frac{\ell_L \ell_R}{\ell_L + \ell_R + 1} \leq \frac{\ell_L + \ell_R}{4} + \frac{1}{3}$$

$$\mu_n \leq \sum_{\ell_1, \dots, \ell_{2^d}} \left(\frac{1}{4} \sum_{j=1}^{2^d} \ell_j + \frac{1}{3} \right) \prod_{j=1}^{2^d} P_{n-1}(\ell_j) = 2^{d-2} \mu_{n-1} + \frac{1}{3}$$

for $d \neq 2$ $\mu_n + C_d'' \leq 2^{d-2} (\mu_{n-1} + C_d'')$

$$\leq 2^{(d-2)n} (\mu_0 + C_d'') \quad \text{with } C_d'' = \{3(2^{d-2} - 1)\}^{-1}$$

for $d = 2$ $\mu_n \leq \frac{n}{3} + \mu_0$

consistent with the log-correction $\mu_n \simeq n/12$

lower bound

mean of the inverse $\eta_n = \sum_{\ell} (\ell + 1)^{-1} P_n(\ell)$

$$u_n = \sum_{\ell_1, \dots, \ell_{2^d-1}} \left(\sum_{j=1}^{2^d-1} \ell_j + 1 \right)^{-1} \prod_{j=1}^{2^d-1} P_n(\ell_j)$$

Jensen's inequality implies $\mu_n \geq \eta_n^{-1} - 1$

recursion $P_n(\ell) = \sum_{\ell_1, \dots, \ell_{2^d}} K(\ell \mid \ell_L, \ell_R) \prod_{j=1}^{2^d} P_{n-1}(\ell_j)$

$$\eta_n = \sum_{\ell_1, \dots, \ell_{2^d}} \left[\sum_{\ell} (\ell + 1)^{-1} K(\ell \mid \ell_L, \ell_R) \right] \prod_{j=1}^{2^d} P_{n-1}(\ell_j) \\ = \tilde{\eta}(\ell_L, \ell_R)$$

$$\tilde{\eta}(\ell_L, \ell_R) = \frac{\ell_L + \ell_R - 1}{(\ell_L + 1)(\ell_R + 1)} = \frac{1}{\ell_L + 1} + \frac{1}{\ell_R + 1} - \frac{1}{(\ell_L + 1)(\ell_R + 1)}$$

recursive identity $\eta_n = 2u_{n-1} - (u_{n-1})^2$

for $d = 1$, where $u_n = \eta_n$, $\eta_n \leq 2\eta_{n-1} \leq 2^n \eta_0 = 2^n / (A_0 + 1)$

$$\mu_n \geq (A_0 + 1) / L_n - 1$$

suggested by ChatGPT 5.2 Pro!!

for $d \geq 3$, we use further clever concavity argument

summary

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